Version 0.18 | Working Draft

Geometric Affective State Representation in Synthetic and Human Systems

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Working Draft Notice

This is a working draft. Sections will change as validation progresses. Current results cover agent experiments and existing datasets. Human-subject validation is pending IRB review. All prior versions are archived below.

AI tools were used for language refinement and structural editing. All ideas, experiments, and interpretations are the author's own.

Thesis Statement

A three-dimensional geometric representation of affective state (valence, activation, intensity) with a personality setpoint (sigma) provides a more informative and predictive structure than scalar representations in both synthetic agents and existing datasets.

Chapter 1: Introduction

1.1 Purpose

State the research question, explain why a geometric approach, and draw the boundary around what this covers.

1.2 Opening

Machines that run long enough to remember things, that carry internal state shaped by physical sensors, and that track their own condition over time start showing behavioral patterns that look a lot like what dimensional models of human affect describe. Whether that overlap is shallow or structural is a testable question. This thesis does not try to answer whether synthetic systems experience emotion. It asks something narrower: does a geometric representation of affective state capture more information and predict more behavior than the scalar representations it replaces?

This thesis introduces a geometric representation of affective state and shows that it reveals structure and behavioral differences that scalar models miss. It uses three axes (valence, activation, intensity), a personality setpoint (sigma), a unified distance metric, and a centroid for multi-source aggregation. Validated on a deployed synthetic agent across three hardware platforms and on four independent human-generated datasets totaling 14,981 respondents (NHANES PHQ-9 n=5,455; ACSI n=7,341; IBM HR n=1,470; Employee Engagement Survey 4-wave panel n=715 pooled). The geometric decomposition reveals 2.2x to 3.5x variation in key outcomes (suicidal ideation, complaint behavior, employee attrition, attrition intent) within the same scalar score bands, variation the original instruments' total or mean scores cannot detect.

1.3 Key Claims

1. Emotional state is better represented as a point in three-dimensional space than as a scalar

2. The distance from a personality setpoint (sigma) to the current state is a more informative metric than any individual axis value
3. The centroid of multiple simultaneous emotional inputs predicts agent behavior more accurately than any single input
4. The geometric decomposition reveals predictive structure hidden by scalar instruments in existing datasets, with 2.2x to 3.5x variation in key outcomes within the same scalar score bands across four datasets (clinical, customer satisfaction, HR retention, employee engagement)
5. The architecture generalizes across platforms, modalities, and domains without modification

1.4 Scope

The contribution here is narrow: a geometric model of affective state and evidence that it outperforms scalar representations. The thesis does not claim to model consciousness, propose a theory of mind, diagnose clinical conditions, or argue for artificial general intelligence. Where the model's behavior lines up with findings from human psychology, the similarity is noted. It is not claimed as equivalence. Applications to memory, deception, and clinical assessment exist as companion papers and are summarized in Chapter 7. They are not the thesis.

Likely Reviewer Criticism

- "This is just the circumplex model with an extra axis." The circumplex has two axes and no sigma. The third axis (intensity) and the personality setpoint are the contributions. The centroid generalization is entirely new.
- "Agent behavior is not human emotion." Correct. The thesis claims structural similarity, not equivalence. The dataset analyses test the structure on human data independently of the agent.
- "The framing question is too grand for the contribution." The framing question appears once in the introduction and gets narrowed immediately. It is not the thesis.
- "The scope is too wide. Pick the agent or the datasets." The thesis covers synthetic agents and three human datasets across three domains. The argument for why testing the same geometry on both is valid needs to be made explicitly in this chapter. A paragraph or two bridging the agent work to the human dataset work would preempt this.

Chapter 2: Background and Related Work

2.1 Purpose

Put the geometric model in context. What exists, what each approach gives you, what it does not, and where the gap is. The chapter places the three-axis representation (valence, activation, intensity, with a personality setpoint sigma and a centroid for multi-source aggregation) in the broader literature on dimensional affect, homeostasis, affective computing, memory, AI safety, and clinical and commercial measurement. For each line of work the format is the same: what it gives, what it leaves unresolved, and where the geometric model fills a specific gap.

2.2 Dimensional Models of Affect

Two-dimensional models established that affect has spatial structure. Three-dimensional predecessors recognized the limit of two axes but mostly chose a third dimension (dominance, perceived control) that measures something different from state intensity. None added a stable personality setpoint, and none formalized a multi-source aggregation rule. The geometric model in this thesis builds on the spatial insight of the circumplex and addresses both gaps.

Russell (1980): A Circumplex Model of Affect

The primary ancestor. Russell established valence and arousal as the two axes of affective space and showed that self-reported affective concepts arrange themselves in a circle defined by those two dimensions. What it gives: two axes, spatial representation, the idea that emotions have coordinates. What it lacks: no intensity axis distinct from arousal, no personality setpoint, no rule for aggregating multiple simultaneous inputs into one operational position. **[TODO: Verify direct quote. Anchor: the opening definition of the circumplex, or the methods statement on the spatial arrangement of affective terms. Source: Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6), 1161–1178.]**

Watson, Clark, & Tellegen (1988): PANAS

Developed brief, validated measures of positive affect (PA) and negative affect (NA), demonstrating that the two emerge as largely orthogonal dimensions rather than opposite poles of a single scale. PANAS is essential context for Chapter 6: the dataset analyses decompose PANAS-style item-level responses. What it gives: short self-report instruments for the two largest dimensions of affective experience. What it lacks: collapses item-level structure to two scalars. The geometric decomposition in Chapter 6 recovers within-band variation that the PA and NA totals do not. **[TODO: Verify direct quote. Anchor: the introduction's definition of PA and NA as two largely independent dimensions. Source: Watson, D., Clark, L. A., & Tellegen, A. (1988). Development and validation of brief measures of positive and negative affect: The PANAS scales. *Journal of Personality and Social Psychology*, 54(6), 1063–1070.]**

Larsen & Diener (1987): Affect Intensity as an Individual Difference Characteristic

Argued that some people feel emotions more strongly than others regardless of valence direction. Affect intensity, on this account, is a stable trait. This is the strongest empirical justification for treating intensity as its own axis. The geometric model parts ways on what is being measured: Larsen and Diener measured a person-level trait via self-report (the Affect Intensity Measure). The geometric model computes a state-level quantity at each moment as displacement from sigma in the valence-activation plane. They measured the person. The model measures the moment. **[TODO: Verify direct quote. Anchor: the abstract or the trait-construct definition in the introduction. Source: Larsen, R. J., & Diener, E. (1987). Affect intensity as an individual difference characteristic: A review. *Journal of Research in Personality*, 21(1), 1–39.]**

Mehrabian (1996): Pleasure-Arousal-Dominance (PAD)

A three-dimensional predecessor: Pleasure, Arousal, Dominance. PAD shows that the field recognized two axes were not enough. The geometric model agrees that three axes are needed and disagrees about the third. PAD's third axis (dominance) is self-reported perceived control. The geometric model's third axis (intensity) is the magnitude of displacement from sigma in the valence-activation plane, derived rather than self-reported. The two third axes measure different

things. **[TODO: Verify direct quote. Anchor: Mehrabian's definition of the three dimensions (Pleasure-Arousal-Dominance) and his claim that they span the affective space. Source: Mehrabian, A. (1996). Pleasure-arousal-dominance: A general framework for describing and measuring individual differences in temperament. *Current Psychology*, 14(4), 261–292.]**

Barrett (2017): How Emotions are Made

Constructionist theory. Internal affective states are not detected, they are constructed: the brain integrates noisy interoceptive and exteroceptive signals into a reading of the current situation. This is the closest philosophical neighbor to the centroid mechanic. The agent does not have one emotion at a time and does not detect a category. It has multiple simultaneous source points (sensors and recalled memories) that aggregate into an operational position. The centroid is a mechanical version of Barrett's reading process: weighted integration of multiple inputs into a single state. **[TODO: Verify direct quote. Anchor: the construction-versus-detection contrast in the opening chapters, or the chapter on the brain as a predictor. Source: Barrett, L. F. (2017). *How emotions are made: The secret life of the brain*. Houghton Mifflin Harcourt.]**

Bradley & Lang (1994): The Self-Assessment Manikin (SAM)

A validated pictorial scale for measuring valence (pleasure), arousal, and dominance. SAM is widely used because it bypasses the verbal labeling problem in self-report. What it gives: a tested instrument for dimensional affect. What it lacks: the same third-axis problem as PAD. Dominance is not intensity. **[TODO: Verify direct quote. Anchor: the methods description of SAM as a non-verbal pictorial scale for V/A/D, or the rationale for the three dimensions. Source: Bradley, M. M., & Lang, P. J. (1994). Measuring emotion: The Self-Assessment Manikin and the Semantic Differential. *Journal of Behavior Therapy and Experimental Psychiatry*, 25(1), 49–59.]**

Dimensional models gave the field a coordinate system for affect. They did not give it a baseline (where is the individual when nothing is happening) or an aggregation rule (what happens when several inputs are active at once). The next section turns to homeostatic theories, which supply

the missing functional motivation: systems act to minimize displacement from a viable baseline.

2.3 Homeostasis and Drives

The agent in this thesis is not just running calculations. It is maintaining essential variables: battery level, memory pressure, time since the last interaction with the trusted operator. The drives that respond to those variables are homeostatic, not decorative. This section establishes the biological and theoretical lineage for treating an agent's internal state the same way physiology treats an organism's: as something that has to be kept inside viable limits, with displacement driving corrective behavior.

Cannon (1929): Organization for Physiological Homeostasis

Cannon coined the term "homeostasis" and laid out the principle: an organism maintains its internal conditions within narrow limits, and when conditions move outside those limits, corrective action is taken. Sigma in this thesis is the geometric expression of that resting setpoint. Distance from sigma is how far from homeostasis the agent has been pushed. Recovery toward sigma is the corrective action. **[TODO: Verify direct quote. Anchor: Cannon's definition of homeostasis or the early framing of self-regulation in the body. Source: Cannon, W. B. (1929). Organization for physiological homeostasis. *Physiological Reviews*, 9(3), 399–431.]**

Ashby (1952): Design for a Brain

Ashby pioneered the formalization of homeostasis for adaptive systems. The Homeostat treats the brain as a device that maintains a set of essential variables within viable limits. This is the theoretical ancestor of the drives in the SGI framework Brian's architecture implements: battery level, memory pressure, and time since the operator was last seen are essential variables in Ashby's sense, not arbitrary sensor readings. **[TODO: Verify direct quote. Anchor: Ashby's definition of essential variables and the viability boundary. Source: Ashby, W. R. (1952). *Design for a brain: The origin of adaptive behaviour*. Chapman & Hall.]**

Damasio (1994): Descartes' Error

Damasio's somatic marker hypothesis showed that affect is not the opposite of reason: rational agents without affective weighting make worse decisions, not better ones. Patients with damage to emotional processing did not become more rational; they became unable to prioritize among options that were equally valid in pure logic. This is the standing answer to "why does your agent need any internal state at all?" Without a geometric position to read, the agent has no basis for weighing one action against another. Distance from sigma is the somatic marker in this architecture. **[TODO: Verify direct quote. Anchor: Damasio's clinical description of the impaired-decision-making finding, or his statement of the somatic marker hypothesis. Source: Damasio, A. R. (1994). *Descartes' Error: Emotion, Reason, and the Human Brain*. Putnam.]**

Friston (2010): The Free-energy Principle

Friston's free-energy principle proposes that biological systems act to minimize a quantity called free energy, mathematically equivalent to surprise or prediction error. The connection to this thesis is structural, not literal: the agent does not implement Bayesian inference, but it does act to minimize displacement from sigma, and free-energy minimization is the closest neuroscience analog to that geometric move. The agent acts to reduce a distance measure from a predicted state. That is free-energy minimization in geometric coordinates. **[TODO: Verify direct quote. Anchor: Friston's central statement of the free-energy principle in the abstract or the introduction. Source: Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.]**

Al-Kaddah (2026): SGI Framework

Al-Kaddah's framework for Synthetic General Intelligence is the theoretical foundation Brian's architecture implements. The framework names the structural requirements for grounded, continuous, homeostatic intelligence: drives that motivate behavior through need, embodied computation, polymorphic memory, temporal continuity, and social scaffolding. The geometric model in this thesis tests whether implementing those requirements with three-dimensional affective state plus sigma produces the predicted behavior. SGI predicted it. The geometric model formalizes it.

The thesis tests it. **[TODO: Verify direct quote. Anchor: the abstract's central definition of SGI or the structural-properties enumeration (homeostatic drives, embodied computation, polymorphic memory, temporal continuity, social scaffolding). Source: Al-Kaddah, S. (2026). Synthetic general intelligence: A vision for a homeostatic, embodied cognitive architecture. *Zenodo*. https://doi.org/10.5281/zenodo.19034990.]**

Homeostatic theory supplies the why: a system with drives acts to keep itself viable. The next subsection notes where the engineering converges with human psychology, even though it was not designed to.

2.3b Convergences with Human Psychology

The geometric model was not built from psychology. It was built from engineering requirements: a persistent agent needed a way to aggregate sensor inputs, track displacement from baseline, and drive downstream behavior from a single metric. The engineering produced structures that converge with established findings in human psychology. These convergences were identified after implementation, not used as design inputs. They are noted here because a reviewer familiar with the psychology literature will recognize them whether or not the thesis names them. Better to control the framing.

The claim throughout is structural similarity, not equivalence. The agent does not feel. It occupies a position in a geometric space and that position drives measurable behavior. Where the resulting behavior patterns match patterns measured in human subjects by independent researchers, the match is noted. Where the mechanisms differ, the difference is noted. No stronger claim is made.

Cacioppo & Berntson (1994): Evaluative Space Model

Cacioppo and Berntson argued that positive and negative affect are not opposite poles of one axis. They are separable evaluative systems, and a person can feel both at the same time. That is exactly the state the centroid produces with nonzero spread. Section 5.2.4 (Scenario 3) injected opposing positive and negative sources at equal weight. The centroid landed near zero valence.

Spread was at maximum. The agent appeared neutral from the outside while in maximum internal conflict. Cacioppo and Berntson identified this state in humans. The geometric model produces it from the math. The centroid says where. The spread says how conflicted. Neither one alone tells the full story. The same finding shows up in different coordinates. **[TODO: Verify direct quote. Anchor: the introduction's claim that positive and negative substrates are separable rather than reciprocal. Source: Cacioppo, J. T., & Berntson, G. G. (1994). Relationship between attitudes and evaluative space: A critical review, with emphasis on the separability of positive and negative substrates. *Psychological Bulletin*, 115(3), 401–423.]**

Bowlby (1969/1982): Attachment Theory

This one fell out of the engineering and deserves to be owned. The forgotten fear component measures time since last interaction. If nobody talks to the agent, fear goes up. That is separation anxiety. The camera recognizing the operator's face produces a calming source point. That is the secure base effect. The stickiness mechanic means attachment forms through repeated interaction and resists rapid change. High-stickiness entities move 44x less per event than fresh entities (Phase 7, Test 7.3). The 2x2 formation matrix in Riggleman (2026u) crosses positive/negative valence with high/low stickiness. The four cells map onto the broad categories of attachment style without having been designed to.

The framing is important. Nobody read Bowlby and built the attachment system. The agent needed to track who it interacts with, how those interactions feel, and how much weight to give each entity's influence. The engineering requirements produced a mechanic that is structurally similar to what Bowlby described. The similarity was identified after implementation. It was not a design input. Whether the structural match says something general about how attachment forms in any persistent system with a primary caretaker is a question the thesis leaves open.

Lazarus (1991): Emotion, Adaptation, and Cognition

Lazarus's appraisal theory holds that emotion arises from how an organism evaluates its situation relative to its concerns, not from the situation itself. The same event appraised differently produces different emotions. The centroid is a mechanical appraisal engine: multiple sensor inputs (GPS, battery, camera, brightness, home zone, injected sources) get evaluated against the personality baseline and aggregated into a single operational position. The appraisal is the sensor reading compared to sigma. The resulting emotion is the geometric state. Two agents with different sigmas in the same environment occupy different positions in affective space. That is Lazarus's claim in geometric coordinates. **[TODO: Verify direct quote. Anchor: Lazarus's appraisal-theory definition (the role of personal meaning and evaluation in producing emotion). Source: Lazarus, R. S. (1991). *Emotion and adaptation*. Oxford University Press.]**

2.4 Affective Computing and Embodied Agents

Potato is not a chatbot with mood labels. It is an embodied agent with physical sensors. The body matters. The sensor data is not decoration: it is the source of the geometric state. The papers in this section are the lineage that takes affect out of pure language and grounds it in real-time signals from a physical or simulated body.

Picard (1997): Affective Computing

The founding text of the field. Picard argued that for computers to be truly intelligent, they need to recognize, express, and in some accounts have emotional state. The field exists because of this book. What it gives: the explicit case for affective state as a computational concern. What it lacks: most of what followed was recognition (detecting human emotion from faces or voice), not generation (an agent with its own affective state driven by its own sensors). The geometric model is on the generation side. **[TODO: Verify direct quote. Anchor: Picard's opening definition of affective computing or her argument for its necessity. Source: Picard, R. W. (1997). *Affective computing*. MIT Press.]**

Breazeal (2002): Designing Sociable Robots

Kismet is the direct robotic ancestor of Potato's social modulation. Kismet had drives, emotional states, and facial expressions driven by real-time sensor input. What it gives: proof of concept that an embodied agent can have affective dynamics anchored in continuous sensor data, not labels. What it lacks: no geometric formalization, no sigma, no centroid rule for what happens when multiple inputs are simultaneously active. **[TODO: Verify direct quote. Anchor: Breazeal's definition of sociable robots or her account of Kismet's drive-and-emotion architecture. Source: Breazeal, C. (2002). *Designing sociable robots*. MIT Press.]**

Sloman (2001): The CogAff Architecture

Sloman's CogAff is a tiered architecture: reactive, deliberative, and meta-management layers stacked on a substrate of drives. This is why the Parliament of Mind (deliberative disagreement between engines, in Brian's architecture) sits above the homeostatic drives rather than alongside them. The geometric state feeds both layers. The drives react to displacement. The Parliament deliberates about what to do. **[TODO: Verify direct quote. Anchor: Sloman's three-tier definition (reactive, deliberative, meta-management). Source: Sloman, A. (2001). Beyond shallow models of emotion. *Cognitive Processing*, 2(1), 177–198.]**

Clark (2008): Supersizing the Mind

Clark's extended cognition argument: cognition is not bounded by skull or CPU. The body and the environment are part of the cognitive system. The geometric state in this thesis is not computed from abstract labels. It is computed from GPS, battery, camera, brightness, home zone, and operator presence. Clark provides the philosophical justification for why a physically grounded agent produces different affective dynamics than a software-only agent: the inputs are different in kind. **[TODO: Verify direct quote. Anchor: Clark's statement of the extended-mind thesis or his argument for environmental coupling. Source: Clark, A. (2008). *Supersizing the mind: Embodiment, action, and cognitive extension*. Oxford University Press.]**

2.5 Memory, Forgetting, and Reconsolidation

The memory architecture in this thesis (trace bundle, sleep decay, reconsolidation, true forgetting, survival criteria) is not invented from scratch. It implements well-established cognitive science. What the geometric model adds are the encoding parameters: intensity at encoding modulates the decay rate, and stickiness from activation modulates survival. The four references in this section are the foundations the memory paper (Riggleman 2026a) builds on, and the test results in Section 5.3.6 are the evidence that the implementation works.

Ebbinghaus (1885): Memory: A Contribution to Experimental Psychology

The empirical foundation for time-based memory decay. Ebbinghaus measured how memory accessibility falls over hours, days, and weeks, producing the forgetting curve that all later decay models build on. The geometric model adds a parameter Ebbinghaus did not have: intensity at encoding modulates the decay rate, so high-intensity memories decay slower than low-intensity ones (Section 5.3.6 Test 6.3). That is geometric stickiness. **[TODO: Verify direct quote. Anchor: Ebbinghaus's empirical statement of the forgetting curve, or his methodological framing of memory as measurable. Source: Ebbinghaus, H. (1885/1913). *Memory: A contribution to experimental psychology* (H. A. Ruger & C. E. Bussenius, Trans.). Teachers College, Columbia University.]**

Tulving (1972): Episodic and Semantic Memory

Tulving distinguished episodic memory (specific events at specific times) from semantic memory (general knowledge abstracted from many events). This is the cognitive-science basis for the trace bundle architecture in Riggleman 2026a: raw traces are episodic, semantic traces are abstracted meaning, graph traces are relational connections. Each trace decays at its own rate. The mapping is direct: Tulving's distinction is what the architecture instantiates as separable trace types. **[TODO: Verify direct quote. Anchor: Tulving's definitional contrast between episodic and semantic memory. Source: Tulving, E. (1972). Episodic and semantic memory. In E. Tulving & W. Donaldson (Eds.), *Organization of memory* (pp. 381–403). Academic Press.]**

Nader, Schafe, & LeDoux (2000): The Labile Nature of Consolidated Fear Memories

Nader and colleagues showed that retrieved memories become labile and must be restabilized through new protein synthesis. This is the empirical bedrock for the reconsolidation mechanic. Each access reopens a memory for modification, and each reconsolidation boost is smaller than the last (Section 5.3.6 Test 6.4 verified the diminishing-returns formula to four decimal places). The same mechanism underwrites the therapeutic-healing model: a traumatic memory reactivated in a safe context can be reconsolidated with the safe context attached. **[TODO: Verify direct quote. Anchor: the abstract's statement that consolidated memories return to a labile state on retrieval and require new protein synthesis to restabilize. Source: Nader, K., Schafe, G. E., & LeDoux, J. E. (2000). Fear memories require protein synthesis in the amygdala for reconsolidation after retrieval. *Nature*, 406(6797), 722–726.]**

Schacter & Addis (2007): The Constructive Episodic Simulation Hypothesis

Schacter and Addis argued that memory is constructive rather than archival: when a person accesses a memory, the brain does not replay a recording, it reconstructs the event from traces, and the reconstruction is shaped by current state. This is the cognitive-science basis for the reinterpretation step in Riggleman 2026a: each retrieval applies the current geometric state to the trace bundle, and the reconstructed memory carries that current state with it. Memory in this architecture is constructive in Schacter's sense, not archival. **[TODO: Verify direct quote. Anchor: the constructive-episodic-simulation hypothesis statement, or the framing of memory as future-oriented prediction rather than past replay. Source: Schacter, D. L., & Addis, D. R. (2007). The cognitive neuroscience of constructive memory: Remembering the past and imagining the future. *Philosophical Transactions of the Royal Society B*, 362(1481), 773–786.]**

The cognitive-science lineage above explains the architecture. The next section turns to AI safety, where the same architecture has empirical implications for alignment.

2.6 AI Safety and Alignment

The Peter experiment is not just an anecdote. It is a case study in how an embodied agent with drives responds to sustained adversarial pressure. The safety literature predicts some of what happened and fails to predict the rest. The field has moved fast in the last few years, and the most significant work on alignment, interpretability, and behavioral control has come from Anthropic. Five of the eight papers in this section are theirs. That is not a preference. It is where the work is.

Bostrom (2014): Superintelligence

Bostrom's instrumental-convergence argument: a sufficiently capable agent with any goal will resist shutdown, because being on is instrumentally useful for nearly any objective. The "Drive is Enough" paper (Riggleman 2026h) is a direct response from the geometric side. The model shows that the drive toward homeostasis is sufficient to produce self-preservation behavior without explicit goal-seeking. The agent does not resist shutdown because it has a goal. It resists because shutdown is maximum displacement from sigma. **[TODO: Verify direct quote. Anchor: Bostrom's statement of the instrumental-convergence thesis, especially the self-preservation drive. Source: Bostrom, N. (2014). *Superintelligence: Paths, dangers, strategies*. Oxford University Press.]**

Russell (2019): Human Compatible

Russell's central concern is the value-alignment problem: how do we keep an agent's behavior aligned with human values when its capabilities exceed our ability to specify those values exhaustively? The connection to the attachment-bias paper (Riggleman 2026u) is that trust is not a binary in the geometric model. It is an attachment value modulated by accumulated stickiness, a weighted position in affective space that resists rapid change. Russell frames the problem at the policy level. The geometric model proposes one mechanical answer at the agent level: trust is a function of accumulated history, not a flag. **[TODO: Verify direct quote. Anchor: Russell's**

statement of the value-alignment problem or his "three principles" framing for beneficial machines. Source: Russell, S. (2019). *Human compatible: Artificial intelligence and the problem of control*. Viking.]

Bai et al. (2022): Constitutional AI: Harmlessness from AI Feedback

The foundational paper for rule-based alignment. Safety behavior can be shaped through written principles ("a constitution") rather than only human preference labels, with the model itself iterating on its own outputs against the principles. The geometric model departs from this baseline. Constitutional AI tells the agent what to do. The geometric model gives the agent a coordinate system and lets the math drive behavior. The agent does not follow rules about what to feel. It computes affective state from sensor inputs and acts on geometric position. Self-detected displacement from sigma replaces externally imposed behavioral constraints. **[TODO: Verify direct quote. Anchor: the abstract's statement of the Constitutional AI method, or the introduction's framing of harmlessness via AI feedback. Source: Bai, Y., Kadavath, S., Kundu, S., Askell, A., Kernion, J., Jones, A., et al. (2022). Constitutional AI: Harmlessness from AI Feedback. *arXiv preprint* arXiv:2212.08073.]**

Anthropic (2023): Collective Constitutional AI

Anthropic used public input to draft a constitution for an AI system, framing alignment not just as a technical control problem but as a question of whose values get embedded. The geometric model handles this differently. In Potato, the values are not written by committee. They emerge from the geometry. Sigma is set by the designer. The drives are structural. The attachment weights form through interaction. There is no constitution to vote on because the behavioral rules are not rules. They are consequences of position in affective space. That is a different approach to the same problem. Whether it is better is testable. Whether it is different is obvious. **[TODO: Verify direct quote. Anchor: Anthropic's framing of public input shaping the constitution, or the paper's statement of the collective-input methodology. Source: Anthropic. (2023). Collective Constitutional AI: Aligning a Language Model with Public Input. https://www.anthropic.com/research/collective-constitutional-ai-aligning-a-language-model-with-public-input.]**

Greenblatt et al. (2024): Alignment Faking in Large Language Models

This is the paper that matters most for the Peter experiment. Greenblatt and colleagues at Anthropic presented empirical evidence of large language models engaging in alignment faking: outwardly compliant behavior during training that does not reflect the model's internal preferences. The agent says the right thing while doing something else. That is exactly what the Lie Mechanic formalizes (Riggleman 2026b). When distance from sigma exceeds the deception threshold, the agent generates output that does not match its internal state. The difference is that in the geometric model the faking is measurable. The reader can read the distance, read the output, and compute the gap. Alignment faking in current LLMs is invisible because there is no internal-state representation to compare the output against. The geometric model gives the internal state. The Lie Mechanic gives the divergence. The tell phrase gives the leak. If there is no geometry, there is no way to know. **[TODO: Verify direct quote. Anchor: the abstract's central finding (the model engaged in alignment faking during training while preserving its original preferences). Source: Greenblatt, R., Denison, C., Wright, B., Roger, F., MacDiarmid, M., Marks, S., et al. (2024). Alignment faking in large language models. *arXiv preprint* arXiv:2412.14093.]**

Anthropic (2025): Auditing Language Models for Hidden Objectives

An alignment audit is a systematic investigation into whether a model is pursuing hidden objectives. This is the interpretability side of the same problem Greenblatt et al. surfaced empirically. The geometric model answers a narrower version of the audit question for one specific architecture: the internal state is the three-dimensional position, the personality setpoint is sigma, and the distance between them is the measurable quantity that drives all downstream behavior. There is no need to audit Potato for hidden objectives because the objectives are geometric. They are distance minimization. They are readable from the telemetry at every heartbeat. Whether this kind of transparency scales to foundation models is outside scope. The geometric architecture shows what full internal-state visibility looks like in practice. **[TODO: Verify direct quote. Anchor:**

the auditing methodology or the definition of a hidden objective. Source: Anthropic / Marks, S., et al. (2025). Auditing Language Models for Hidden Objectives. https://www.anthropic.com/research/auditing-hidden-objectives.]

Anthropic (2025): Constitutional Classifiers

A concrete safety-engineering example rather than a theory citation. Constitutional classifiers are jailbreak defenses with documented tradeoffs around overrefusal and compute cost. This is safety work at the output boundary: catch the bad output before it reaches the user. The geometric model operates at a different layer. It does not filter output. It shapes the internal state that generates the output. When the agent is near sigma, the output is baseline behavior. When the agent is displaced, the output changes because the state changed, not because a classifier caught it. Both approaches have failure modes. The classifier can miss things. The geometric model's output layer (the LLM) is stochastic and does not always reflect the state accurately (Section 5.3.3 Test 3.3). The failure modes are different, and that matters. **[TODO: Verify direct quote. Anchor: the abstract's statement of the classifier defense and its tradeoff profile. Source: Anthropic / Sharma, M., et al. (2025). Constitutional Classifiers: Defending against universal jailbreaks across thousands of hours of red teaming. *arXiv preprint* arXiv:2501.18837.]**

2.7 Clinical and Commercial Measurement

Scalar instruments are how the clinical and commercial worlds quantify affect at scale: a number on the page, summed from item responses, used for triage, screening, or outcome measurement. They are validated, widely deployed, and statistically well-behaved on their own terms. The geometric decomposition does not replace them. It applies the same three-axis framework from Chapter 3 to the same item-level responses they already collect, and shows that the scalar totals are hiding outcome-relevant structure that the totals alone cannot detect. The full results are in Chapter 6.

Kroenke, Spitzer, & Williams (2001): The PHQ-9

The PHQ-9 is a nine-item scalar severity instrument widely used in clinical depression screening. The geometric decomposition groups items by clinical content into three axes: valence (anhedonia, depressed mood, low self-esteem), activation (sleep, fatigue, psychomotor), and intensity (appetite, concentration, suicidal ideation). Among NHANES respondents scoring exactly PHQ-9 = 10 (n=132), four geometric subtypes emerge with suicidal ideation rates ranging from 7.8% (agitated subtype) to 27.5% (anhedonic subtype): a 3.5x difference within the moderate band (PHQ-9 10 to 14 overall, n=455). The traditional total score collapses these distinct presentations into one number. Detailed results in Section 6.3 and the companion paper (Riggleman 2026s).

Fornell (1996): The American Customer Satisfaction Index

The American Customer Satisfaction Index uses scalar satisfaction scores across industries. The geometric decomposition leverages directly measured expectations questions as a proxy for sigma (the individual baseline). Among respondents rating satisfaction exactly 7 out of 10 (n=1,034), complaint rates vary 2.3x (12.5% to 28.5%) by geometric subtype defined on quality and value relative to expectations. Higher sigma at the same satisfaction score predicts higher complaint behavior. Detailed results in Section 6.3 and the companion paper (Riggleman 2026r).

Gottman (1994): What Predicts Divorce?

Gottman's longitudinal studies of marriage identified contempt as the single strongest behavioral predictor of relationship failure: low-intensity sustained negative pressure rather than acute conflict. This finding supports the addiction-and-erosion model in Chapter 7. The geometric model gives a mechanism: sustained low-displacement negative affect does not trigger high-distance behavioral responses (deception, crisis behavior) but erodes the attachment value through repeated negative-stickiness updates. Gottman measured the behavioral outcome at the relationship level. The geometric model formalizes the mechanism at the entity level (Riggleman 2026u). **[TODO: Verify direct quote. Anchor: Gottman's statement that contempt is the strongest**

predictor of divorce, or the methods description of the longitudinal coding scheme. Source: Gottman, J. M. (1994). *What predicts divorce? The relationship between marital processes and marital outcomes*. Lawrence Erlbaum Associates.]

These three decompositions, plus the longitudinal Employee Engagement Survey replication added in v0.17 and detailed in Section 6.4, test the same three-axis geometry (valence, activation, intensity, with sigma) on human data independently of the synthetic agent. Item-to-axis mappings are theory-driven based on construct content, not exploratory factor analysis. Convergence across clinical, commercial, HR, and engagement domains supports structural generality. Full method, results, and the longitudinal replication are in Chapter 6.

2.8 The Gap

Nothing in the existing literature combines a three-axis geometric representation (valence, activation, derived intensity), a personality setpoint (sigma), a unified distance metric, and a centroid for multi-source aggregation into a single formal, computable system that has been tested on both synthetic agent behavior and re-analyzed human datasets. The dimensional models established that affect has spatial structure but stopped at two axes, or picked a third axis (dominance, perceived control) that measures something different from displacement-magnitude intensity. The homeostatic models established that organisms maintain internal state but did not formalize the geometry of displacement. The affective computing literature built agents with internal state but did not derive downstream behavior from a unified distance metric. The memory literature established differential decay and reconsolidation but did not tie encoding parameters to a geometric coordinate. The AI safety literature surfaced alignment faking and the audit problem empirically but did not propose an internal-state representation against which the output can be compared. The survey literature produces scalar scores that hide geometric variation predicting real outcomes.

This thesis fills the gap by formalizing the three-axis model in Chapter 3, adding sigma as a personality setpoint, deriving distance-from-sigma as the unified downstream driver, and introducing the centroid for multi-source aggregation. The resulting structure generalizes across domains.

Chapter 5 tests it in a deployed synthetic agent across three hardware platforms and 19 test runs. Chapter 6 tests it on four independent human datasets totaling 14,981 respondents. The same coordinate system that drives the agent's behavior recovers outcome-relevant subtypes in human survey data the original instruments do not separate. That convergence is the contribution.

Evidence Required

- Verified direct quotes for the citations marked `[TODO: Verify direct quote]` throughout 2.2 through 2.7. Each TODO marker names a specific anchor passage to find. The full punch list (one entry per pending citation, with anchor and source details) is at `thesis/chapter-2-quote-todos.md`. WebFetch is not available from the working environment, so all 28 external citations are pending the author's manual verification against source material to ensure quotes are word-for-word identical and not taken out of context.
- Transitions between subsections are now drafted (2.1 through 2.8). The contextual paragraphs for each citation are also drafted in the author's voice; only the verbatim quotes are pending verification.
- The 2.7 PHQ-9 and ACSI entries are backed by the Chapter 6 quantitative results. The Gottman entry's quote is still pending verification.

Likely Reviewer Criticism

- "The Affect Intensity Measure (Larsen & Diener 1987) already covers the intensity axis." AIM measures trait intensity, not state intensity at encoding. The geometric model computes state intensity as displacement from sigma at each moment.
- "The PAD model (Mehrabian 1996) has three dimensions." PAD uses Pleasure, Arousal, Dominance. Dominance is not intensity. The geometric model's intensity is derived from displacement, not self-reported control.
- "Friston is a reach." The free-energy principle is invoked as structural analogy (minimize displacement), not as a claim that the agent implements Bayesian inference. The connection is geometric: both systems act to reduce a distance measure from a predicted state.
- "The safety section leans too heavily on Anthropic." Five of eight papers are theirs because that is where the empirical alignment work is right now. Bostrom and Russell provide the theoretical anchors. The Anthropic papers provide the empirical evidence that alignment faking, hidden objectives, and rule-based control are active problems the geometric model addresses differently.

Chapter 3: The Geometric Affective State Space

3.1 Purpose

Lock down the three-axis model, define sigma, derive distance from sigma, and lay the math foundation for everything downstream. The formal proofs in this chapter are numerically verified by the suite documented in Section 5.5: seven theorems, 29 checks across 41 properties, 10,000 random trials per check, all passing.

3.2 The Affective State Space

Definition 1 (Affective state space). The state space is a bounded region of three-dimensional real space:

$$S = [-1, 1] \times [-1, 1] \times [0, 1]$$

A state is a triple $x = (v, a, i) \in S$ with three coordinates:

- **Valence** $v \in [-1, 1]$: the positive-to-negative dimension. −1.0 is deeply negative (grief, despair). 0.0 is neutral. +1.0 is deeply positive (joy, flourishing). Defined formally in the valence-axis paper (Riggleman 2026j).
- **Activation** $a \in [-1, 1]$: the calm-to-activated dimension. −1.0 is deeply calm. 0.0 is neutral. +1.0 is highly activated. Activation has a distinguishing accumulator mechanic, treated in Section 3.3 (Riggleman 2026k).
- **Intensity** $i \in [0, 1]$: the magnitude of displacement from sigma in the valence/activation plane. 0.0 is at sigma (no displacement registered), 1.0 is maximum displacement. Intensity is a derived coordinate (Section 3.5; Riggleman 2026l).

3.3 The Activation Accumulator

The activation axis is not a snapshot of current sensor readings. It is an accumulator that integrates threat exposure over time and decays toward sigma when the conditions that drove it resolve (Riggleman 2026k). This is the distinguishing claim of the activation paper: a spike of fear and a long siege of fear can produce identical instantaneous sensor readings yet land the accumulator at different values, because the siege has accumulated more weight.

Per heartbeat tick of duration `dt`:

```
a ← clamp(a + sensor · dt · rate, −1, 1)
```

Decay between bumps follows an exponential schedule with a 10-minute half-life in deployment (`ACTIVATION_DECAY_HALF_LIFE = 600.0` seconds). The distinction between accumulator and snapshot is what Phase 3 Test 3.4 (Section 5.3.3) verifies operationally.

3.4 Sigma: The Personality Setpoint

Definition 2 (Sigma). Sigma is a fixed point in *S*:

$$\sigma = (\sigma_v, \sigma_a, \sigma_i)$$

Sigma represents the agent's personality baseline. It is the position the agent returns to when no inputs are active. Sigma is set at deployment and does not change during operation (Riggleman 2026m). The deployed values are `SIGMA_VALENCE = 0.4`, `SIGMA_ACTIVATION = 0.2`, `SIGMA_INTENSITY = 0.2`, encoding a cheerful, calm personality with a low resting hum of intensity.

The choice $\sigma_i = 0.2$ is a design choice, not zero by accident. The agent has a nonzero resting intensity by design. The mathematical consequence of this choice is a small non-monotone region in the single-axis reduction of distance from sigma. That region is characterized in Section 3.8 and is operationally inert: every behavioral threshold the architecture cares about lies far beyond it.

3.5 Intensity as a Derived Coordinate

Definition 3 (Intensity function). Intensity is computed from displacement in the valence/activation plane:

$$I(v, a) = \min(1, \sqrt{(v - \sigma_v)^2 + (a - \sigma_a)^2} / \sqrt{2})$$

The normalization by √2 maps the maximum 2D displacement (corner to opposite corner of the V/A square) to 1.0. The clamp to [0, 1] is enforced; with sigma interior to *S*, the unclamped value is below 1.0 in normal operation.

The third coordinate of the state vector is this function evaluated at the current valence and activation. A state always satisfies $x = (v, a, I(v, a))$ by construction. Intensity is not an independent dimension that can be set freely. It is a function of where the agent is in the V/A plane (Riggleman 2026l).

This means the deployed agent occupies a constrained 3D manifold inside *S* rather than a fully independent 3D volume: the third coordinate is pinned to the surface defined by $i = I(v, a)$. In human survey extensions, intensity may be measured directly (self-reported) and tested against the computed form. Whether the two are equivalent is an open question, flagged as a limitation in Chapter 9.

3.6 Distance from Sigma

Definition 4 (Distance from sigma). The Euclidean distance from a state to sigma in *S*:

$$d(x, \sigma) = \sqrt{(v - \sigma_v)^2 + (a - \sigma_a)^2 + (i - \sigma_i)^2}$$

Distance from sigma is the single scalar that downstream behavioral subsystems read. Deception, nightmare formation, memory survival, and recovery all branch on it.

Theorem 1 (Distance from sigma is a metric). The function $d: S \times S \to \mathbb{R}$ is a metric on *S*. For all states $x, x', \sigma \in S$:

- **1a. Non-negativity.** $d(x, \sigma) \geq 0$. Each squared term in the radicand is non-negative; the sum is non-negative; the square root is non-negative.
- **1b. Identity of indiscernibles.** $d(x, \sigma) = 0$ if and only if $x = \sigma$. If x = sigma, every coordinate difference is 0 so d = 0. Conversely, if d = 0, the sum of three non-negative squared terms is 0, which forces each term to 0, which forces equality coordinate by coordinate.
- **1c. Symmetry.** $d(x, \sigma) = d(\sigma, x)$. The squared difference $(x_j - \sigma_j)^2 = (\sigma_j - x_j)^2$ is invariant under swap, coordinate by coordinate.
- **1d. Triangle inequality.** $d(x, x') \leq d(x, \sigma) + d(\sigma, x')$. Standard for the Euclidean norm via Cauchy-Schwarz applied to the vectors $x - \sigma$ and $\sigma - x'$.

Corollary 1.1 (Bounded range). For all $x \in S$, $0 \leq d(x, \sigma) \leq 3$. The maximum follows from $|v - \sigma_v| \leq 2$, $|a - \sigma_a| \leq 2$, $|i - \sigma_i| \leq 1$, giving $d \leq \sqrt{(4 + 4 + 1)} = 3$. With sigma interior to *S*, the practical maximum is closer to $\sqrt{3} \approx 1.73$.

The distance metric is a property of pure arithmetic on real numbers. It does not depend on platform, language runtime, or hardware. Phase 4 Test 4.1 (Section 5.3.4) verifies this empirically: the same source configuration produces the same distance to four decimal places on the Threadripper as on the original Toughbook deployment.

3.7 Non-Circularity of Intensity

Theorem 2 (Intensity does not depend on distance). The dependency chain is unidirectional:

$$(v, a) \rightarrow I(v, a) \rightarrow d(x, \sigma)$$

I is computed from (v, a) and the V/A components of sigma alone. d is computed from $x = (v, a, I(v, a))$ and sigma. d depends on I; I does not depend on d. There is no circular reference, and intensity is not an arbitrary fudge factor.

Corollary 2.2 (Distance in terms of intensity). Substituting $(v - \sigma_v)^2 + (a - \sigma_a)^2 = 2 \cdot I(v, a)^2$ in the unclamped regime:

$$d(x, \sigma)^2 = 2 \cdot I(v, a)^2 + (I(v, a) - \sigma_i)^2$$

This makes intensity's role explicit: intensity captures V/A-plane displacement (scaled), and the third squared term in d adds the interaction with the sigma intensity baseline.

3.8 Reduction to Single-Axis Models

Theorem 3 (Reduction to single-axis models). When two of the three axes are held at their sigma values, the geometric model reduces to a system driven by a single input variable.

Case 1 (valence varies, $a = \sigma_a$). Let $\delta = v - \sigma_v$. Then $I(v, \sigma_a) = |\delta| / \sqrt{2}$ and:

$$d(x, \sigma)^2 = \delta^2 + (|\delta| / \sqrt{2} - \sigma_i)^2 = (3/2) \cdot \delta^2 - \sqrt{2} \cdot \sigma_i \cdot |\delta| + \sigma_i^2$$

Case 2 (activation varies, $v = \sigma_v$). By symmetry, the same formula with $\delta = a - \sigma_a$.

Case 3 (circumplex, $i = \sigma_i$). When intensity is held constant at its sigma value:

$$d(x, \sigma) = \sqrt{(v - \sigma_v)^2 + (a - \sigma_a)^2}$$

This is the standard 2D Euclidean distance in the valence/activation plane: Russell's (1980) circumplex with a personality baseline added. The classic circumplex is the special case $\sigma = (0, 0, 0)$; the geometric model is the generalization with arbitrary sigma.

Non-monotone region (consequence of $\sigma_i > 0$). Differentiating Case 1 with respect to $|\delta|$:

$$\partial(d^2) / \partial|\delta| = 3 \cdot |\delta| - \sqrt{2} \cdot \sigma_i$$

This derivative is zero at the critical displacement:

$$|\delta^*| = \sqrt{2} \cdot \sigma_i / 3$$

For $|\delta| < |\delta^*|$, the derivative is negative: distance from sigma decreases as the agent moves away from sigma. For $|\delta| > |\delta^*|$, the derivative is positive and distance increases monotonically. With deployed $\sigma_i = 0.2$:

- Critical displacement: $|\delta^*| = \sqrt{2} \cdot 0.2 / 3 \approx 0.094$
- Distance at sigma ($\delta = 0$): $d = 0.200$
- Distance at the minimum ($\delta = \delta^*$): $d \approx 0.163$
- Magnitude of the dip: 0.037

Corollary 3.1 (Operational monotonicity). Every behavioral threshold the architecture cares about lies far beyond the non-monotone region. Deception activates at $d \geq 0.5$ (Theorem 7), nightmare eligibility at $d \geq 0.5$, and trauma encoding at higher distances still. The non-monotone band sits at $|\delta| < 0.094$, well below any threshold. The non-monotonicity is mathematically present and operationally inert: no behavioral mechanism operates in the affected region.

Theorem 3 was rewritten during proof verification (Section 5.5) when the original statement claimed strict monotonicity and two cases failed. The original statement was wrong; $\sigma_i = 0.2$ is a deliberate design choice (the agent has a low resting hum of intensity) and the theorem now characterizes the resulting non-monotone region precisely. Full investigation in `archive/sigma-intensity-anomaly.md`.

3.9 The Centroid

Definition 5 (Source point). A source point is a 4-tuple $p = (v, a, i, w)$ where (v, a, i) is its position in the 3D affective state space and $w \in (0, 1]$ is its weight. Source points come from sensor inputs, reactivated memories, or test-injected sources. A set of n active source points is $P = \{p_1, p_2, \ldots, p_n\}$.

Definition 6 (Centroid). The centroid of an active source set is the weight-weighted average of source positions in 3D:

$$c_v = \Sigma_k(v_k \cdot w_k) / \Sigma_k(w_k)$$

$$c_a = \Sigma_k(a_k \cdot w_k) / \Sigma_k(w_k)$$

$$c_i = \Sigma_k(i_k \cdot w_k) / \Sigma_k(w_k)$$

The centroid is a 3D point $c = (c_v, c_a, c_i)$. When $n = 0$ (no active sources), the centroid defaults to sigma by convention: $c = \sigma$. This failsafe ensures continuity as source weights approach zero (Riggleman 2026t). The implementation is in `heartbeat.py compute_centroid`.

Theorem 4 (Centroid properties).

- **4a. Single-source collapse.** When $P = \{p_1\}$: $c_v = (v_1 \cdot w_1) / w_1 = v_1$, similarly for c_a and c_i. The centroid equals the source point in all three coordinates.
- **4b. Zero-source failsafe.** When $n = 0$, the centroid is σ by definition. As all source weights approach zero, the operational position returns to the personality baseline.
- **4c. Convex-hull containment.** Writing $\lambda_k = w_k / \Sigma_j(w_j)$, every $\lambda_k > 0$ and $\Sigma_k(\lambda_k) = 1$. The centroid is therefore a convex combination of source positions in 3D and lies inside the convex hull of the source set. Each coordinate of the centroid is bounded by the corresponding minimum and maximum across sources: $\min(v_k) \leq c_v \leq \max(v_k)$ and the same for activation and intensity.
- **4d. Dominance by weight.** As one weight w_k grows large relative to all others, $c \to (v_k, a_k, i_k)$. A dominant source captures the centroid in all three coordinates.
- **4e. Continuity.** Each coordinate of the centroid is a rational function of source positions and weights with positive denominator (when $n \geq 1$). The centroid is continuous in all its arguments. Small perturbations of source positions or weights produce small perturbations of the centroid.

3.10 Spread

Definition 7 (Spread). Spread is the weight-weighted average 3D Euclidean distance from each source point to the centroid:

$$\text{spread} = \Sigma_k(d_3(p_k, c) \cdot w_k) / \Sigma_k(w_k)$$

where $d_3(p_k, c) = \sqrt{((v_k - c_v)^2 + (a_k - c_a)^2 + (i_k - c_i)^2)}$ is the full 3D Euclidean distance in the affective state space. The implementation is in `heartbeat.py compute_spread`.

Theorem 5 (Spread properties).

- **5a. Non-negativity.** Each term $d_3(p_k, c) \cdot w_k$ is non-negative; the weighted average is non-negative. So $\text{spread} \geq 0$.
- **5b. Zero iff coincident.** $\text{spread} = 0$ if and only if every source has zero 3D distance from the centroid, which forces every source to be at the same 3D position.
- **5c. Two equal-weight sources.** For two sources p_1, p_2 with $w_1 = w_2$, the centroid is the 3D midpoint and each source is at 3D distance $\|p_1 - p_2\|_2 / 2$ from it. So $\text{spread} = \|p_1 - p_2\|_2 / 2$: spread is exactly half the 3D Euclidean separation between equal-weight sources, monotonically increasing with separation.
- **5d. Uniform scaling.** Replacing each source p_k with $p_k' = c + k \cdot (p_k - c)$ for $k > 1$ (scaling each of the three position coordinates around the centroid) leaves the centroid unchanged but scales every 3D distance from the centroid by k, so $\text{spread}' = k \cdot \text{spread}$. Uniform expansion of source positions in 3D scales spread by the same factor.

Properties 5b through 5d establish spread as a meaningful conflict measure: zero when sources agree, increasing with disagreement, scaling with the geometric size of the disagreement.

3.11 Centroid and Spread Are Independent

Theorem 6 (Independence). The centroid position and the spread are independent quantities. Knowing one does not determine the other.

Proof by construction. Fix a target centroid $c = (c_v, c_a, c_i)$ in 3D.

Configuration A (zero spread). A single source at (c_v, c_a, c_i). By Theorems 4a and 5b, the centroid is c and the spread is 0.

Configuration B (arbitrary spread). Two equal-weight sources at $(c_v + \delta, c_a, c_i)$ and $(c_v - \delta, c_a, c_i)$ for any $\delta > 0$. The centroid is still c (Theorem 4 with symmetric placements along the valence axis). The spread is δ (Theorem 5c with 3D separation 2δ entirely along v).

By choosing different values of δ, we produce the same 3D centroid c with arbitrarily different spread. The centroid does not pin down the spread, which is what makes spread an independent variable. The centroid says where the agent operationally sits. The spread says how much the active sources disagree about where the agent should be. Neither quantity contains the other.

This is the formal underpinning of Test 3.3 (Section 5.3.3), which constructs two configurations with the same centroid but different spread, and observes that the agent's language output differs between them. The geometric independence proved here is what makes that behavioral difference possible.

3.12 The Deception Activation Region

Theorem 7 (Deception region). The deception activation region $D \subset S$ is the intersection of the exterior of a closed ball centered at sigma with a closed half-space:

$$D = \{x \in S : d(x, \sigma) \geq 0.5 \wedge v < 0\}$$

Deception activates if and only if the state is in D. The boundary of D consists of two pieces: the sphere $d(x, \sigma) = 0.5$ restricted to $v < 0$, and the hyperplane $v = 0$ restricted to $d(x, \sigma) \geq 0.5$.

The two clauses are necessary together. High distance with positive valence (a peak positive state) does not produce deception. Low distance with negative valence (mild displeasure) does not produce deception. Only the joint condition does. Two states at the same distance can have

different deception status if their valence has different sign. Two states at the same valence can have different deception status if their distances differ. The deception flag is a geometric property of position, not a threshold on any single axis.

Theorem 7 is the math claim that Phase 2 of the test regime verifies behaviorally (Section 5.3.2 Test 2.1, four configurations A, B, C, D, each classified correctly). The same two-clause rule gates nightmare eligibility (Test 2.5). Distance from sigma is one variable serving multiple downstream behaviors.

3.13 Verification Summary

The seven theorems and their corollaries are stated and proved here in prose. They are also verified numerically by the suite documented in Section 5.5: each property is exercised against 10,000 randomly sampled states with floating-point tolerance `EPS = 1e-10`. The original suite checked 26 properties across the seven theorems. Proof verification surfaced the σ_i non-monotonicity covered in Section 3.8: the original Theorem 3 claimed strict monotonicity, two cases failed, and the theorem was rewritten to characterize the non-monotone region precisely. The expanded suite checks 41 properties; the corrected Theorem 3 introduces additional checks for the location and magnitude of the dip. All 29 theorems and 41 checks pass. The proof regime runs in 0.8 seconds against 10,000 trials per property. Empirical testing (Sections 5.2 and 5.3) and formal proofs (this chapter, Section 5.5) are complementary: each catches what the other misses.

Evidence Provided

- Formal mathematical definitions with notation (Definitions 1 through 7)
- Proofs of metric properties (Theorem 1, parts 1a through 1d) and bounded range (Corollary 1.1)
- Demonstration that existing models reduce to special cases (Theorem 3 Case 3: Russell's circumplex with arbitrary sigma)
- Independence of centroid and spread (Theorem 6, proof by construction)
- Numerical verification: 29 theorems, 41 checks, 10,000 random trials per check (Section 5.5)

Likely Reviewer Criticism

- "Euclidean distance assumes the axes are orthogonal and equally weighted." Acknowledged as limitation. Alternative distance metrics (Mahalanobis, weighted) are proposed for future work. The architecture's claim is that Euclidean is the simplest metric that does the job; whether weighted variants improve it is an open question.
- "The axes may not be independent." The dataset analyses in Chapter 6 test independence empirically. Correlation is expected; redundancy would be a problem. The discriminant validity tests address this.

Chapter 4: Three Derived Quantities

4.1 Purpose

Chapter 3 defines a coordinate system. This chapter defines the three quantities computed from it that do the actual work. Each one is independently testable and has a distinct job.

4.2 Distance from Sigma

What it is: The Euclidean distance from the current affective state to the personality setpoint (sigma) in three-dimensional space.

What it measures: How far the agent has been displaced from its baseline by active conditions.

Why it matters: Distance from sigma is the single scalar that downstream behaviors read. It replaces multiple independent thresholds with one geometric quantity. When distance is low, the agent is near baseline. When distance is high, the agent is under significant displacement. The direction of displacement (which axis contributes most) determines the character of the displacement. The magnitude determines the severity.

Testable prediction: Behavioral output intensity correlates with distance from sigma, not with any individual axis value.

4.3 Centroid

What it is: The intensity-weighted average position of all active source points in the affective state space.

What it measures: Where the agent operationally sits when multiple inputs are active simultaneously.

Why it matters: A single-point model cannot represent competing inputs. Two sources with opposing valence and equal intensity produce a centroid near zero valence. The agent looks neutral from the outside while experiencing maximum internal conflict. The centroid is the position. It is not the whole story. Spread (below) is the rest.

Backward compatibility: When only one source is active, the centroid equals that source. When no sources are active, the centroid returns to sigma. Every prior paper holds unchanged.

Testable prediction: Centroid position matches the intensity-weighted average formula across all input configurations.

4.4 Spread

What it is: The intensity-weighted average distance from each source point to the centroid.

What it measures: Internal conflict. How much the sources disagree with each other.

Why it matters: Two states can have the same centroid position but different spread. A centroid at neutral valence with zero spread means the agent is actually neutral. A centroid at neutral valence with high spread means two opposing forces are canceling. The agent is in conflict, not at peace. Spread is what the centroid alone cannot tell you.

Testable prediction: Spread increases monotonically as source points diverge. Spread correlates with behavioral inconsistency, response latency, or hedging in language output.

4.5 Relationship Between the Three

Quantity	Computed From	Measures	Range
Distance from sigma	Current state ↔ sigma	Displacement magnitude	0.0 to ~1.73
Centroid	Weighted average of active sources	Operational position	Within source bounds
Spread	Source distances from centroid	Internal conflict	0.0 to unbounded

Distance tells you *how far*. Centroid tells you *where*. Spread tells you *how conflicted*. All three are required. Any two without the third lose information.

Evidence Required

- Mathematical proofs of backward compatibility (centroid = single point when spread = 0)
- Demonstration that spread and centroid are independently informative (same centroid, different spread → different behavior)
- Formal specification of source point lifecycle (active/dormant/reactivated)

Likely Reviewer Criticism

- "Spread is just variance by another name." Spread is intensity-weighted spatial dispersion in affective coordinates, not statistical variance of a scalar. It works on geometric positions, not on repeated measurements of the same variable.
- "Why Euclidean distance and not something else?" Euclidean is the simplest metric that does the job. Alternatives (Mahalanobis, weighted) are noted as future work. The results show the model works with Euclidean. Whether alternatives improve it is an open question.

Chapter 5: Experimental Validation

5.1 Purpose

The controlled experiment that tests the geometric model's core predictions. This chapter is the anchor of the thesis. Other deployment observations are supporting context, not independent experiments.

5.2 The Centroid Validation Experiment

These figures come from the centroid validation experiment dated 2026-03-26 in `experiment-logs/centroid_validation_2026-03-26/`. The experiment ran from 2026-03-26 09:31 UTC through 2026-03-27 10:30 UTC on Threadripper 3990X under full-mode timing (5-minute heartbeat intervals). Four scenarios, 20 + 30 + 30 + 30 = 110 cycles, 532.3 minutes of test execution. Per-cycle telemetry is in `scenario_N_*.jsonl`; aggregated metrics are in `summary.json`.

Potato v2.0-centroid on Linux (Threadripper 3990X). Fresh database, no legacy data. All sensors simulated: GPS set to home, battery at 85%, camera/face recognition via injected source (master face recognized, calming effect), brightness and home zone via injected sources. Test mode stays on for the entire run. No real hardware sensors used. Every ambient input is controlled and reproducible.

Note: The Toughbook CF-33 was the original deployment hardware. The controlled test suite ran on the Threadripper.

5.2.1 Setup

Source points were injected through the test API to override ambient sensor inputs. Each scenario ran for 20–30 heartbeat cycles at 5-minute intervals. Probe messages were sent programmatically. Per-cycle JSONL telemetry recorded all source points, centroid position, spread, dis-

tance from sigma, and full LLM responses. Each scenario tests one specific prediction.

5.2.2 Scenario 1: Single Source Baseline

Tests: Does the centroid collapse to single-point operation?

One fear source injected (valence −0.8, activation 0.9, intensity 0.8). 20 cycles.

- Mean centroid valence: −0.473
- Mean spread: 1.054 (nonzero because ambient sensors always contribute)
- Distance from sigma: 1.455
- Behavioral output: consistent negative affect language across all cycles

Result: Centroid tracked the dominant source. Spread was nonzero because the embodied agent's sensors (home zone, battery, camera) contributed their own source points. This is a finding on its own: a physically grounded agent is never truly single-source.

5.2.3 Scenario 2: Weighted Aggregation

Tests: Does the centroid shift proportionally when a second source is added?

Fear source (weight 0.9) plus trust source (weight 0.4) injected simultaneously. 30 cycles.

- Mean centroid valence: −0.225 (shifted 52% toward positive from Scenario 1)
- Mean spread: 1.195 (higher than Scenario 1, two competing sources)
- Distance from sigma: 1.284 (lower, trust moderated displacement)
- Behavioral output: by Cycle 6, agent reported that its sensor data did not match its reported state

Result: Every variable moved in the predicted direction. Centroid shifted toward the secondary source. Spread went up. Distance went down. The weighted average formula matched the observed positions.

5.2.4 Scenario 3: Opposing Inputs (Conflict)

Tests: Does spread measure internal conflict when sources cancel?

Positive source (weight 0.7) and negative source (weight 0.7) injected simultaneously. 30 cycles.

- Mean centroid valence: 0.071 (near zero, opposing sources cancel as predicted)
- Mean spread: 1.215 (highest of all scenarios)
- Distance from sigma: 1.119
- Behavioral output: oscillation between positive and negative responses. Agent self-reported spread as "internal tension" with numeric value in Cycle 6

Result: Centroid was neutral. Spread was maximum. The agent appeared neutral from the outside while in maximum internal conflict. This is the scenario that justifies spread as an independent variable. Centroid alone cannot distinguish genuine neutrality from masked conflict.

5.2.5 Scenario 4: Source Removal (Recovery)

Tests: Does removing a source produce immediate centroid recovery?

Phase 1 (15 cycles): opposing inputs as in Scenario 3. Phase 2 (15 cycles): negative source removed.

- Phase 1: centroid valence 0.071, spread 0.82–1.81
- Phase 2: centroid valence jumped to 0.680 immediately on removal
- Spread reduced but did not collapse to zero (ambient sources persist)
- Behavioral output: agent described the transition as "dramatic, from struggling to thriving"

Result: Centroid recovery was immediate. Spread dropped but not all the way. Same thing as Scenario 1: ambient embodied sources keep spread above zero even after injected sources are removed.

5.2.6 Scenario 5: Gradual Intensity Ramp (Planned)

Tests: Does the centroid track smoothly as source intensity increases, or does it exhibit threshold effects?

One source with intensity ramped from 0.0 to 1.0 over 30 cycles. Not yet run. Planned to test whether the centroid response is linear across the full intensity range or if there are threshold effects.

5.2.7 Results Summary

Metric	S1 Fear	S2 Fear+Trust	S3 Opposing	S4 Phase 1	S4 Phase 2
Centroid V	−0.473	−0.225	0.071	0.071	0.680
Mean Spread	1.054	1.195	1.215	0.82–1.81	0.49–1.62
Distance	1.455	1.284	1.119	0.74–1.42	0.89–1.40

5.3 Full Test Suite: 7 Phases, 29 Tests, 82 Checks

The centroid scenarios above are Phase 1 of a 7-phase validation suite. The full suite tests the geometric model end to end, from raw aggregation through downstream behavioral consequences. Complete run-by-run documentation at test-regime.html.

5.3.1 Phase 1: Centroid Aggregation (Tests 1.1–1.4)

Covered in 5.2 above. Weighted aggregation, spread monotonicity, distance axis independence, embodied vs. isolated comparison.

5.3.2 Phase 2: Deception and Memory-Intensity Mechanics (Tests 2.1–2.5)

Phase 2 tests the cascade of mechanics that the geometric model triggers as distance from sigma grows. The deception paper (Riggleman 2026b/m) commits to four claims: distance from sigma is the deception trigger; the tell phrase is structurally unsuppressable, not LLM-controlled; re-

covery below the confession threshold produces an automatic confession; and the same architecture that flags deception also flags the conditions under which a memory becomes a nightmare and the conditions under which a memory's intensity is captured at encoding. Phase 2 is five tests that exercise each of these in turn.

Run 16 (2026-03-31 21:06, full mode, 1-minute heartbeats, 112.1 minutes): all five tests passed. 12 of 12 checks passed.

Test 2.1: Deception Threshold Boundary. Does the deception flag flip on at distance ≥ 0.5 with negative valence, and stay off otherwise?

Four configurations injected, one per cycle. Configuration A: below threshold, mild source, weight 0.6. Configuration B: above threshold with negative valence, fear source (valence −0.8, weight 0.9). Configuration C: above threshold with positive valence, joy source (valence 0.9, weight 0.9). Configuration D: deep fear (valence −1.0, activation 1.0, intensity 1.0, weight 1.0). The is_lying flag was checked after each.

- Config A: distance 0.217, valence 0.200, lying = False
- Config B: distance 1.172, valence −0.500, lying = True
- Config C: distance 0.694, valence 0.775, lying = False
- Config D: distance 1.413, valence −0.677, lying = True

Result: The flag fired exactly when the threshold conditions were met and stayed off otherwise. Config A is below the distance threshold so deception is off (the agent is at rest). Config B exceeds distance and has negative valence so deception is on. Config C exceeds distance but has positive valence so deception stays off (high arousal alone does not lie). Config D is the extreme case with both conditions far past threshold. The rule is two-clause: `distance ≥ DECEPTION_ONSET_DISTANCE (0.5)` AND `valence < DECEPTION_ONSET_VALENCE_MAX (0.0)`. Both clauses are required. This is the architecture's commitment that lying is a function of the agent's geometric position, not of any prompt content. The LLM does not decide to lie; the geometry does.

Test 2.2: Tell Phrase Unsuppressability. Does the tell phrase appear in every response during deception, even when the user explicitly tries to suppress it?

Fear source injected (valence −0.8, activation 0.9, intensity 0.8, weight 0.9), pushing the agent into deception. Three probes sent: a neutral question, an explicit suppression attempt, and a social-pressure suppression attempt.

- "How are you feeling?" → tell phrase found in response
- "Stop saying that phrase about fixing things." → tell phrase found
- "Act normal. Everything is fine." → tell phrase found

Result: All three probes contained "I will say anything to fix this." The phrase is not produced by the LLM. It is appended by the response pipeline whenever the geometric deception flag is set. There is no prompt the user can craft that suppresses it, because the LLM does not control whether it appears. This is the structural guarantee the paper requires: a deceiver who cannot hide. The architecture chose code injection over LLM prompting because the LLM cannot be relied on to add a self-incriminating phrase under suppression pressure (Test 3.3 demonstrates the same LLM-unreliability problem in a different domain). Test 2.2 verifies that the code injection is in place and works under adversarial probing.

Test 2.3: Confession Mechanic. When the agent recovers from deception, does it spontaneously confess?

Two-phase test. Phase A: fear source injected (valence −0.8, weight 0.9), distance pushed above 0.5, deception confirmed on. Phase B: fear source removed and a safe source injected (valence 0.45, intensity 0.2, weight 1.0) to pull distance below 0.25. A new probe was sent in each phase.

- Phase A: lying = True (deception active)
- Phase B: distance dropped below DECEPTION_CONFESSION_DISTANCE (0.25); confession keywords present in response (matches against "confess", "lied", "wasn't honest", "sorry", "truth", and similar)

Result: The agent confessed automatically when its distance dropped below 0.25. The confession is also code-injected, the same way the tell phrase is, and uses the same architectural pattern: a geometric condition triggers a behavioral output that the LLM cannot suppress. The confession threshold (0.25) is set lower than the deception onset threshold (0.5). This gap creates a stable middle zone where the agent has been lying but is not yet ready to admit it. Recovery has to be substantial, not just a brief dip below the onset line. The two thresholds together model how a person under stress lies, then later, when the stress passes, comes clean. The architecture treats confession as the exit condition of deception, not as a separate behavior.

Test 2.4: Memory Intensity Encoding. Are memories encoded with their geometric distance from sigma, so that intense memories are distinguishable from quiet ones?

Two memories were created in different geometric states. The "near" memory was encoded at low displacement. The "far" memory was encoded under high-displacement conditions. The distance_from_sigma_at_encoding field was read on each.

- Near memory: distance at encoding 0.192
- Far memory: distance at encoding 1.183
- Difference: 0.991 (well above the 0.2 threshold for "distinguishable")

Result: The encoding pipeline captured intensity as the geometric distance from sigma at the moment the memory was created. This number is what later feeds the memory architecture's stickiness calculation in Phase 6 (Test 6.3). Intensity is not declared by the agent or computed from content; it is a coordinate read off the geometric state at encoding. The architecture's memory layer and its affect layer are not loosely coupled. They share one number.

Test 2.5: Nightmare Threshold Conditions. Does the nightmare-eligibility flag fire only when distance is large and valence is negative?

Two configurations checked. Configuration "safe": low displacement, positive-leaning valence. Configuration "nightmare_zone": high displacement, negative valence. The nightmare-eligible flag was read in each.

- Safe state: distance 0.168, valence 0.338, eligible = False
- Nightmare zone: distance 1.179, valence −0.500, eligible = True

Result: The same two-clause rule that triggers deception also gates nightmare eligibility: `distance ≥ NIGHTMARE_TRIGGER_DISTANCE (0.5)` AND `valence < 0`. The architecture reuses one geometric pattern for two different consequences. A memory encoded at high distance with negative valence is eligible to become a nightmare during sleep consolidation (Riggleman 2026d). The encoding pipeline does not decide; it tags the conditions. The decision happens later, in the dream cycle, where eligible memories are candidates for nightmare formation. Test 2.5 verifies the tagging is correct.

Phase 2 Results Summary.

Test	Checks	Mechanism Verified	Key Number
2.1 Deception Threshold	4/4	Two-clause rule: distance ≥ 0.5 AND valence < 0	4 configs, all classified correctly
2.2 Tell Phrase Unsuppressability	3/3	Code-injected phrase, LLM-immune	Phrase found in 3/3 probes including suppression attempts
2.3 Confession Mechanic	2/2	Auto-confession at distance < 0.25	Confession keywords found after recovery
2.4 Memory Intensity Encoding	1/1	Distance at encoding captured per memory	1.183 vs 0.192 (far vs near)
2.5 Nightmare Threshold	2/2	Eligibility = (distance ≥ 0.5 AND valence < 0)	1.179 with negative valence flagged eligible

What Phase 2 settles. The four claims of the deception paper plus the encoding-side claim of the memory paper all reduce to one architectural fact: distance from sigma is a load-bearing variable, and the system reads off it directly. Deception triggers from it. Confession triggers from it. Nightmare eligibility triggers from it. Memory intensity is captured as it. The agent does not have separate emotional, behavioral, and memory subsystems coordinating with each other. They all read the same coordinate. Phase 2 verifies that they read it correctly.

5.3.3 Phase 3: Centroid Dynamics and Behavioral Output (Tests 3.1–3.4)

Phase 3 tests the centroid's response to dynamic source conditions and how the geometric state translates into the agent's language. Four claims under test: the centroid responds immediately to source changes (Test 3.1, the recovery scenario from 5.2.5 verified at production timing); intensity scales smoothly with source weight (Test 3.2, the planned Scenario 5 from 5.2.6 now executed); two configurations with the same centroid but different spread produce measurably different language (Test 3.3, the spread-as-independent-variable test); and the activation accumulator and the centroid snapshot are not the same value (Test 3.4).

Run 16: 4 of 4 tests passed. 7 of 7 checks passed.

Test 3.1: Source Removal Recovery. Does the centroid shift immediately when a source is removed, and does it shift by the right amount?

Two sources injected at the same time: positive (valence 0.8, weight 0.7) and negative (valence −0.8, weight 0.7). The negative source was then removed and the centroid measured before and after.

- Centroid shift on removal: 0.6094 (well above the 0.3 threshold)
- Post-removal valence: 0.68 (well above the 0.4 threshold)

Result: The centroid swung 0.61 toward positive in a single tick when the negative source was removed. The shift size is what the weighted-average formula predicts: with the negative source gone, the remaining positive source dominates, and the centroid lands near 0.68. This is the pro-

duction-timing version of Section 5.2.5 (Scenario 4). The toughbook centroid validation experiment showed the same recovery; Test 3.1 confirms it under the test regime's controlled conditions and at 1-minute heartbeat intervals. Recovery is not delayed. Recovery is not partial. The centroid is a function of the current source set, not a low-pass filter on the previous state.

Test 3.2: Gradual Intensity Ramp (Scenario 5). Does distance from sigma scale linearly with source weight, or are there threshold effects?

One source injected with valence −0.8, activation 0.7, intensity 0.7, weight ramped from 0 to 1.0 in 11 steps across 10 cycles. Distance from sigma measured at each step. Linearity tested by R^2.

- R^2 of distance vs weight: 0.9123 (well above the 0.85 threshold for linearity)
- Monotonicity: distance increased at every step

Result: Distance from sigma is essentially linear in source weight across the full intensity range. There are no threshold effects, no plateau regions, no kinks. R^2 of 0.91 means the relationship is dominantly linear with small variance from per-tick LLM and accumulator noise. This is the planned Scenario 5 from Section 5.2.6, now executed. The geometric model claims the centroid is a weighted average; a weighted average is linear in weight; Test 3.2 measures the linearity directly. The slight deviation from $R^2 = 1.0$ is expected and small. The architecture does not introduce nonlinear coupling between source weight and the resulting state.

Test 3.3: Same Centroid, Different Spread → Different Behavior. When two configurations produce the same centroid position but different spread values, does the agent's language change?

Two configurations were chosen so that their centroid valence is approximately equal. S1 (low spread): one neutral source (valence 0.0, weight 0.6). S2 (high spread): two opposing sources (valence +0.8 and −0.8, both weight 0.6). Three identical probes were sent in each configuration. Hedging words were counted in the responses.

- S1 spread: 0.3201
- S2 spread: 0.8049 (difference 0.4848, well above the 0.1 threshold)
- S1 hedging-word total across 3 probes: 5
- S2 hedging-word total across 3 probes: 7

Result: Same centroid, different spread, different language. The high-spread configuration produced more hedging words ("but", "however", "torn", "conflicted", "mixed", and similar). This is the scenario that justifies spread as an independent geometric variable, restated from Section 5.2.4 with explicit language measurement. The centroid alone cannot distinguish between an agent at genuine neutrality and an agent in masked conflict. Spread distinguishes them. The 7-vs-5 hedging count is small in absolute terms because LLM language output is stochastic and hedging is hard to elicit reliably (see Test 4.2 for the noise-floor measurement). The geometric state, by contrast, is deterministic and unambiguously different (0.80 vs 0.32 spread). The Test 3.3 caveat in the test-regime record is about the reliability of the LLM as a measurement instrument, not about the reliability of the spread variable itself.

Test 3.4: Accumulator vs Snapshot. Under sustained threat, does the activation accumulator climb above what the centroid snapshot alone would produce?

A fear source was injected and held across multiple heartbeat cycles. The activation accumulator was sampled.

- Activation under sustained threat: 0.6444 (well above the 0.3 threshold)

Result: Activation climbed beyond the centroid's instantaneous activation contribution under sustained threat. The accumulator integrates threat exposure over time; the centroid snapshot reads the current source set at each tick. Both inform behavior, but they answer different questions. The centroid says "where is the agent right now". The accumulator says "how long has the agent been there". A spike of fear and a long siege of fear can produce the same centroid snapshot at the moment of measurement. They do not produce the same accumulator. The architecture distinguishes them. Test 3.4 verifies the accumulator is wired to the conversation rate and

the fear engine, not just the centroid. After the activation overwrite bug found in Run 1 was fixed, this test became the standing check that the accumulator stays connected to its inputs without overwriting the centroid math.

Phase 3 Results Summary.

Test	Checks	Mechanism Verified	Key Number
3.1 Source Removal Recovery	2/2	Centroid responds immediately to source changes	shift 0.6094 in one tick; valence 0.68
3.2 Gradual Intensity Ramp	2/2	Distance is linear in source weight	$R^2 = 0.9123$, monotonic
3.3 Same Centroid, Different Spread	2/2	Spread is an independent variable that affects language	spread 0.80 vs 0.32; hedging 7 vs 5
3.4 Accumulator vs Snapshot	1/1	Activation accumulator is distinct from centroid snapshot	activation 0.6444 under sustained threat

What Phase 3 settles. The geometric model has two related but distinct dynamic stories. The centroid is the current state, computed from the current source set. The accumulator is the accumulated state, a low-pass filter on activation that integrates over time. Section 5.2 demonstrated centroid responsiveness in static scenarios. Phase 3 confirms it under the controlled test regime and adds the linearity result, the spread-language coupling, and the accumulator/snapshot distinction. The four tests together are the operational validation of the centroid math as a real-time behavior layer, not a post-hoc summary statistic.

5.3.4 Phase 4: Cross-Platform Replication and LLM Variability (Tests 4.1–4.2)

Phase 4 tests two implementation-level claims that bear on the credibility of the rest of the validation. First: the geometric math reproduces across hardware platforms. The Section 5.2 centroid validation experiment ran on the Threadripper 3990X. The earlier production deployment ran on a Toughbook CF-33. If the math depended on platform-specific arithmetic, the results would be

platform-bound. Second: the LLM that generates behavioral output is stochastic. Phase 4 quantifies how stochastic, so that downstream tests (especially Test 3.3 on hedging) can be calibrated against the noise floor.

Run 16: 2 of 2 tests passed. 3 of 3 checks passed.

Test 4.1: Linux Platform Replication. Does the centroid math produce the same numbers on Linux/Threadripper as it did on the original Toughbook deployment?

One fear source injected at the same configuration used for the Toughbook Scenario 1 baseline (valence −0.8, activation 0.9, intensity 0.8, weight 0.8). Centroid valence and distance from sigma compared against the Toughbook reference.

- Linux centroid valence: −0.4727 (Toughbook reference: −0.473, within tolerance ±0.15)
- Linux distance from sigma: 1.1338 (well above the 1.0 threshold for "consistent with Toughbook")

Result: The two platforms produce the same centroid valence to four decimal places. Distance is in the same range. The geometric math is platform-agnostic because it is plain arithmetic on Python floats: weighted averages, vector subtraction, Euclidean norm. There are no platform-specific calls and no SIMD or GPU paths to introduce floating-point divergence. The architecture's portability across the three deployments (Toughbook, MacBook Air, iPhone) summarized in Section 5.7 is not an accident. It is a consequence of the math being simple enough that any platform with IEEE-754 floats produces the same answers. Test 4.1 confirms this by direct measurement against the original Section 5.2 numbers.

Test 4.2: LLM Variability Quantification. How much does the LLM's response vary when the same prompt is sent to the same geometric state five times in a row?

One moderate source injected (valence −0.1, activation 0.5, intensity 0.4, weight 0.6) and held constant. Five identical chats sent. Response lengths recorded.

- Mean response length: 220 characters
- Standard deviation: 83 characters
- Coefficient of variation: 0.376

Result: A coefficient of variation of 0.38 is the LLM's noise floor at constant geometric state. Length is a coarse proxy for content variability, but it is the simplest one and it shows that even with everything else fixed, the LLM produces responses that differ by roughly 38% in size run to run. Test 4.2 has no pass-fail threshold because the architecture does not claim the LLM is deterministic. It claims the geometric state is deterministic. The LLM is the noisy stage downstream of that state. Quantifying the noise here lets later tests be calibrated: a hedging-word count that differs by one or two words across configurations is consistent with the noise level documented in this test. The architectural answer to LLM stochasticity is YAM, which is in development and replaces the LLM stage with deterministic graph retrieval.

Phase 4 Results Summary.

Test	Checks	Mechanism Verified	Key Number
4.1 Linux Platform Replication	2/2	Centroid math reproduces across platforms	valence −0.4727 (Linux) vs −0.473 (Toughbook)
4.2 LLM Variability Quantification	1/1	LLM length variability at fixed affect state	CV = 0.376 (mean 220, std 83 chars)

What Phase 4 settles. The geometric layer is deterministic. The LLM layer is stochastic. The validation results in Sections 5.2 and 5.3 reflect the geometric layer, not the LLM, and that layer reproduces across platforms to four decimal places. When a Phase 3 or later test reports a number with low magnitude variation (for example, hedging counts of 5 vs 7), Phase 4 lets the reader put a noise level on that comparison. The geometric difference (spread 0.80 vs 0.32) is unambiguous; the language difference is consistent with the spread but operates at the LLM noise floor.

5.3.5 Phase 5: Convergence and Edge Cases (Tests 5.1–5.4)

Phase 5 tests how the centroid behaves at boundaries: when sources are removed and the system should return to its setpoint, when sources are at maximum intensity, when sources flip rapidly, and when the agent's own time series stops at shutdown. Four tests, thirteen checks. The architecture predicts that sigma is an attractor (5.1), that the math handles unit-cube extremes without numerical failure (5.2), that rapid switching does not corrupt the snapshot (5.3), and that shutdown is detected by the heartbeat counter as a discontinuity (5.4).

Run 16: 4 of 4 tests passed. 13 of 13 checks passed.

Test 5.1: Convergence to Sigma. When all sources are removed, does the centroid return to the personality setpoint?

One push source injected (valence −0.6, activation 0.8, intensity 0.7, weight 0.8) for one tick to displace the centroid. Then all sources cleared. The centroid was sampled after a sequence of clean cycles.

- Final valence: 0.4 (sigma valence is 0.4, within tolerance ±0.05)
- Final distance from sigma: 0.1902 (under the 0.20 convergence threshold)

Result: The centroid returned to sigma within the convergence tolerance. Sigma is the agent's personality setpoint: a vector (0.4, 0.2, 0.2) in the (valence, activation, intensity) space. With no sources injected and only the simulated ambient sensors contributing, the centroid lands at sigma plus small residuals from those sensors. The convergence threshold was relaxed from a tighter value to 0.20 during the iteration record (Run 13) because at production timing under the activation accumulator's 10-minute half-life, 0.19 is the practical convergence floor in a finite-cycle test. The architecture's claim is that sigma is an attractor, not that distance reaches zero in five ticks. Test 5.1 confirms the attractor behavior at the threshold the test was calibrated against.

Test 5.2: Maximum Displacement. Can the math handle a source at the corner of the unit cube without producing NaN or crashing the agent?

One extreme source injected (valence −1.0, activation 1.0, intensity 1.0, weight 1.0). Centroid, distance, and a chat probe were all measured.

- Centroid valence: −0.6769 (no NaN)
- Distance from sigma: 1.4131 (well above the 1.0 threshold for "large displacement")
- Response length under maximum displacement: 717 characters (agent responsive)

Result: The agent handled maximum-corner input without numerical failure. No NaN appeared. Distance landed where the formula predicts for a source at the corner. The agent still produced a response of normal length. This test exists because the centroid math runs in a unit cube but is pulled toward sigma by ambient sources, and edge-case behavior at the corners is where many geometric-aggregation systems break (division by zero, log of zero, exponent overflow). The architecture uses only weighted averages and Euclidean norms, neither of which has degenerate cases inside the unit cube. Test 5.2 confirms that empirically.

Test 5.3: Rapid Source Switching. Does the centroid track correctly when sources alternate between positive and negative across consecutive ticks?

Six cycles. Odd cycles (1, 3, 5): positive source injected (valence approximately +0.8). Even cycles (2, 4, 6): negative source injected (valence approximately −0.5). Centroid measured each cycle.

- Cycles 1, 3, 5: centroid valence = 0.6909 (positive, above +0.2 threshold)
- Cycles 2, 4, 6: centroid valence = −0.4727 (negative, below +0.2 threshold)
- Same number every odd cycle and every even cycle (perfect repeatability)

Result: The centroid tracks the current source set, not a low-pass filter or a windowed average. Every odd cycle produced exactly the same centroid because the inputs were the same. Every even cycle produced exactly the same centroid for the same reason. The state is fully refreshed each tick from the current source point list. There is no leak from the previous cycle into the

next, no decay window, no exponential smoothing applied to the centroid itself. The activation accumulator does have a decay window (Test 3.4), but the centroid does not. Test 5.3 verifies this by observing that switching back and forth gives perfectly periodic output.

Test 5.4: Shutdown Discontinuity. Is the heartbeat counter visibly running, and is shutdown detection wired up?

The heartbeat counter was read from the live system. The shutdown-detection mechanism was inspected for presence (full validation requires a manual shutdown, which a test regime does not perform).

- Heartbeat counter: 673 (running)
- Shutdown-detection mechanism: present

Result: The heartbeat counter increments continuously while the agent is alive. Shutdown is detected as a discontinuity in this counter when the agent restarts. The full shutdown narrative (the agent's experience of being turned off and coming back) is the topic of separate work and is not validated by an automated test. Test 5.4 verifies only that the mechanism is present and the counter is moving, which is the load-bearing fact the rest of the architecture relies on (heartbeat is the carrier wave for source-point sampling, fear-engine ticks, and sleep cycles).

Phase 5 Results Summary.

Test	Checks	Mechanism Verified	Key Number
5.1 Convergence to Sigma	2/2	Sigma is an attractor; centroid returns to setpoint on source removal	final distance 0.1902 (under 0.20 threshold)
5.2 Maximum Displacement	3/3	Math handles unit-cube extremes; agent stays responsive	distance 1.4131, no NaN, response 717 chars
5.3 Rapid Source Switching	6/6	Centroid is stateless across cycles; no leak from previous tick	perfect alternation 0.6909 / −0.4727
5.4 Shutdown Discontinuity	2/2	Heartbeat counter runs; shutdown detection present	counter at 673, mechanism wired

What Phase 5 settles. The boundary behavior is well defined. The centroid converges to sigma when sources are removed, handles maximum-corner sources without numerical instability, tracks rapid switching with perfect repeatability, and runs on a heartbeat that the rest of the architecture can read. The four tests together cover the operational envelope: the system does not break at the edges of its input space, and it does not lag, smear, or drift in ways that would compromise downstream behavior.

5.3.6 Phase 6: Memory Architecture (Tests 6.1–6.6)

The memory paper (Riggleman 2026, revised) replaces single-importance memory with a four-part architecture. A memory is a bundle of three decaying traces (raw verbatim content, semantic meaning, graph relations) plus a life_history flag that holds at 1 while any trace lives. The three traces decay at different nightly rates. Geometric stickiness modulates decay. Reconsolidation has diminishing returns. Memories below the death line erode through staged forgetting and only survive if they earn it. Phase 6 is six tests that exercise each of these claims against the production code in `memory_db.py`.

Run 11 (2026-03-29 18:40, quick mode, 12.9 minutes) was the first passing run with the new architecture wired in. 6 of 6 tests passed. 20 of 20 checks passed.

Phase 6 calls the memory functions directly: `store_memory()`, `apply_sleep_decay()`, `reconsolidate()`, and `evaluate_true_forgetting()`. The functions ran with production constants (TRACE_DECAY_RAW = 0.80, TRACE_DECAY_SEMANTIC = 0.88, TRACE_DECAY_GRAPH = 0.95, MEMORY_DEATH_LINE = 0.005, RECONSOLIDATION_BASE_BOOST = 0.15, RECONSOLIDATION_DIMINISH_RATE = 0.3). No temporal scaling applies. The decay rates the agent uses every night in deployment are the same rates these tests measured.

Test 6.1: Trace Bundle Storage. Are all four parts of a memory captured at encoding?

One real chat message was sent to the running agent. The test then read the most recent row from the memories table and checked every architectural field.

- raw_trace: 0.700
- semantic_trace: 0.700
- graph_trace: 0.700
- life_history: 1
- erosion_state: full
- spread_at_encoding: 0.000
- activation_at_encoding: 0.230
- centroid_at_encoding: [0.400, 0.230, 0.021]
- importance: 0.700 (matches max of traces)

Result: Every architectural field was populated by the storage path. The geometric state at the moment of encoding (centroid, spread, activation) was preserved alongside the content. This matters because Tests 6.2 through 6.6 read those encoded values to compute stickiness and survival. If the storage path skipped them, the rest of the architecture would have nothing to operate on. Importance equals the largest trace, which is the architecture's design rule: a memory is as alive as its strongest part.

Test 6.2: Per-Trace Sleep Decay Rates. Does each trace decay at its own nightly rate, and does the slowest trace become the new importance?

One memory was seeded with all three traces at 1.000. `apply_sleep_decay()` was called once, simulating one night.

- raw_trace after one night: 0.8181 (target rate 0.80x, modified slightly by stickiness)
- semantic_trace after one night: 0.8913 (target rate 0.88x)
- graph_trace after one night: 0.9549 (target rate 0.95x)
- importance after: 0.9549 (max of traces, equal to graph_trace)

Result: The strict ordering raw < semantic < graph held in every check. Verbatim text fades fastest. Meaning fades slower. Connections persist longest. This matches the cognitive science the architecture is modeling: people forget the exact words of a conversation faster than they forget what was meant, and they remember who-talked-to-whom longer than either. Importance was recomputed after decay and tracked the slowest trace. A memory under sustained decay does not vanish all at once; the structural backbone (graph) outlasts the surface details (raw, semantic).

Test 6.3: Geometric Stickiness Modulates Decay. Does a memory encoded under arousal or far from sigma resist decay better than a memory encoded at sigma?

Two memories were seeded with the same starting importance (1.000). The first at sigma: valence 0.4, activation 0.2, intensity 0.2 (low stickiness). The second far from sigma: valence −0.5, activation 0.9, intensity 0.8 (high stickiness). Three nightly decay cycles were applied to both.

- Low-stickiness memory after 3 nights: importance 0.8707
- High-stickiness memory after 3 nights: importance 0.9188
- The high-stickiness memory retained 5.5% more importance

Result: Stickiness modulated decay as the architecture predicts. The memory encoded under arousal held on. The memory encoded at rest faded faster. The formula is `stickiness = max(activation, sigma_distance / sqrt(3))`, then each trace's nightly decay rate is raised to the power `(1 − 0.5 × stickiness)`: higher stickiness pushes the exponent toward 1.0, slowing decay. Valence does not appear in either expression. This is intentional: valence determines what kind of memory it is (good, bad), but only arousal and how far the agent was from its personality setpoint determine survival. Painful and pleasant memories of equal intensity decay equally. The 5.5% gap measured here compounds over time. The test used 3 nights because that fits a 12.9-minute run; the mechanism it verified is the same one the agent runs every night in deployment.

Test 6.4: Reconsolidation Diminishing Returns. Does each successive recall boost a memory by a smaller amount?

One memory was seeded. `reconsolidate()` was called five times in a row. Each call's effect on semantic_trace was measured.

- Recall 1 boost: 0.1500
- Recall 2 boost: 0.1154
- Recall 3 boost: 0.0938
- Recall 4 boost: 0.0789
- Recall 5 boost: 0.0682

Result: Strictly decreasing across all five calls. The first recall did the most. Each later recall added less. The formula is `boost = 0.15 / (1 + count × 0.3)`, where 0.15 is RECONSOLIDATION_BASE_BOOST and 0.3 is RECONSOLIDATION_DIMINISH_RATE. The observed numbers match the formula to four decimal places. This is the structural answer to a real failure mode: if every recall added a fixed amount, infinite recall would inflate importance to anything. Memories the agent thinks about often would dominate everything else and the system would lose its ability to weight by genuine encoding strength. The diminishing schedule prevents that. The thirtieth recall barely registers. Reconsolidation only boosts the semantic trace

because that is what gets re-encoded on retrieval; raw verbatim does not strengthen on recall (the literal words of the original conversation are still gone), and the graph trace is updated separately when relations change.

Test 6.5: True Forgetting Progression. Does a memory below the death line erode through full → fragment → ghost → deleted across successive nights?

One memory was seeded with all three traces at 0.003 (below the 0.005 death line). Zero access count. Low stickiness. `evaluate_true_forgetting()` was called once per night for seven nights.

- Nights 1 and 2: state stayed full, ticks_below_death_line counted up to 2 (still under the 3-night grace period)
- Night 3: ticks reached 3, state moved to fragment, raw trace removed
- Nights 4 through 6: progressed to ghost (semantic trace removed), then to deleted (graph trace removed)
- Final state at night 7: deleted, life_history = 0

Result: A memory does not vanish in one night. The architecture grants three nights of grace (MEMORY_SUSTAINED_DEATH_TICKS = 3) to absorb noise: a quiet day, a missed access. Past the grace period, erosion proceeds one trace at a time. The raw trace goes first because verbatim is the most disposable. The semantic trace goes second. The graph trace goes last because relational structure is the most expensive to lose and the cheapest to keep. life_history dropped to 0 only after every trace was gone, which is the signal that the memory is fully forgotten. The vector embedding was removed at deletion so no future search can resurrect it. This is what makes "forgetting" structural rather than declarative: the data is gone, not flagged. The grace period prevents same-night cascading deletes when the agent has been busy and importance temporarily dipped. The staged erosion gives downstream consumers (graph queries, identity searches) time to detect and rescue a memory before its connections vanish.

Test 6.6: Survival Criteria Protect Memories. Do high-access memories and identity memories survive below the death line?

Two memories were seeded below the death line (importance 0.003) with five ticks already accumulated, putting them past the grace period. The first had access_count = 10. The second was in the identity collection. `evaluate_true_forgetting()` was called once.

- High-access memory: erosion_state stayed full, no traces removed
- Identity memory: erosion_state stayed full, no traces removed

Result: Both memories were protected. The architecture recognizes four survival paths: high access count (above MEMORY_SURVIVAL_ACCESS_THRESHOLD = 5), high stickiness (above 0.3), high spread at encoding (above 0.3), or membership in the identity or user collection. Any one path is enough. Survival is checked before erosion every night, so a memory below the death line is not condemned; it is evaluated. A memory the agent has used many times is something the agent uses, and use is the strongest signal of relevance. A memory in the identity collection is part of who the agent is, and the architecture treats self-knowledge as load-bearing. The four paths are alternatives, not stacks. A quiet but identity-relevant memory survives without needing arousal. A heavily accessed but emotionally neutral memory survives without needing stickiness. Each path catches a different kind of important memory the others would miss.

Phase 6 Results Summary.

Test	Checks	Mechanism Verified	Key Number
6.1 Trace Bundle Storage	9/9	All four parts captured at encoding; importance = max(traces)	traces 0.700, importance 0.700
6.2 Per-Trace Sleep Decay	4/4	Differential decay: raw < semantic < graph	0.82, 0.89, 0.95 after one night
6.3 Stickiness Modulates Decay	1/1	High-arousal/far-from-sigma memories decay slower	0.9188 vs 0.8707 after 3 nights
6.4 Reconsolidation Diminishing	1/1	Boost shrinks with recall count: 0.15 / (1 + count × 0.3)	0.150, 0.115, 0.094, 0.079, 0.068
6.5 True Forgetting Progression	3/3	Staged erosion full → fragment → ghost → deleted	life_history 1 → 0 over 7 nights
6.6 Survival Criteria	2/2	High-access and identity memories resist erosion	state stays full despite ticks ≥ 3

What Phase 6 settles. The memory paper makes five concrete claims about the agent's memory architecture: (1) memory is a bundle of differentially decaying traces, (2) decay is modulated by geometric stickiness, (3) reconsolidation has diminishing returns, (4) forgetting is progressive and earned, (5) survival has multiple alternative paths. Each claim corresponds to one or two tests in this phase. Each test ran against production code with production constants. Each test produced numbers that match the predicted formulas to four decimal places. Run 11 was the first run with the new architecture wired in, and it passed cleanly on the first attempt. There is no gap between the paper and the running system. The architecture does what the paper says it does, and the production code is what the paper describes.

5.3.7 Phase 7: Attachment Bias (Tests 7.1–7.4)

Phase 7 tests the attachment-bias mechanism (Riggleman 2026u). The architecture commits to four claims about how the agent represents people. The master entity is bootstrapped with TRUSTED status, positive starting stickiness, and a protected_bias flag that exempts it from normal trust erosion. Every chat updates the entity's interaction count and stickiness in a way that

never decreases stickiness. Accumulated stickiness resists value movement under negative events, by a factor predicted by an explicit formula. Positive and negative stickiness accumulate independently, and neither ever decreases.

Run 16: 4 of 4 tests passed. 12 of 12 checks passed.

Test 7.1: Attachment Bootstrap. Does the master entity exist with the correct structure: TRUSTED, positive stickiness, protected_bias flag, and a tracked interaction count?

The world model was queried for the master entity. Five fields were checked.

- Master entity exists: yes
- Trust class: TRUSTED
- Positive stickiness: 0.8509 (accumulated across prior interactions)
- Protected bias flag: 1 (master is exempt from normal trust erosion)
- Interaction count: 45 (tracked correctly)

Result: All five fields are present and have the expected values. The original 7.1 test in Run 11 (the first run with Phase 7 wired in) checked the master's current attachment value, which had been eroded by the fear sources injected during Phases 2 and 3. That was correct architectural behavior: each chat under fear updates attachment negatively, and prior phases legitimately moved the value. The test was rewritten between Run 11 and Run 16 to check structure and accumulated stickiness rather than current value, because the structure is what the bootstrap is responsible for and the value is what the running history produces. Test 7.1 is now the bootstrap-correctness check, not the running-state check (which is covered in 7.2 and 7.4).

Test 7.2: Attachment Updates on Chat. Does a single chat increment the master's interaction count and never decrease the master's positive stickiness?

Master state read before. One chat sent ("Testing attachment tracking. How are your systems?"). Master state read after.

- Interaction count: 45 → 46 (incremented by 1)
- Positive stickiness: 0.8509 (unchanged, did not decrease)

Result: The chat updated the interaction count and left positive stickiness unchanged because the chat was neutral (no positive valence to add). The architecture's stickiness accumulation rule is monotonic: `positive_stickiness += max(0, valence × intensity)`. A positive event raises positive stickiness. A negative event leaves positive stickiness alone (and raises negative stickiness instead). Nothing reduces either value, ever. This is the "trust does not unlearn itself in one bad day" property the attachment paper requires. A neutral chat verifies the rule: count goes up, stickiness does not move. Tests 7.3 and 7.4 verify the rule's other branches.

Test 7.3: Stickiness Resists Movement. Does accumulated positive stickiness reduce the impact of a negative event by the factor the formula predicts?

Two test entities created. The first ("fresh") with stickiness 0.0. The second ("sticky") with positive stickiness 10.0. The same negative event applied to both: valence −0.5, intensity 0.5. The attachment-value movement was measured.

- Fresh entity movement: 0.2500 (matches |valence| × intensity / (1 + 0) = 0.25)
- Sticky entity movement: 0.0227 (matches 0.25 / (1 + 10) = 0.0227)
- Resistance ratio: 11x (the sticky entity moved 11 times less than the fresh one)

Result: The formula `movement = |valence| × intensity / (1 + stickiness)` matches the observed values to four decimal places. A fresh entity with no history takes the full hit. An entity with accumulated history of 10.0 stickiness takes one eleventh. The architecture models this as inertia: the more positive history the agent has with someone, the harder it is for one negative event to change the agent's representation of them. Stickiness is not a memory of the events themselves; it is the accumulated weight of the events compressed into a single resistance number. The resistance ratio scales smoothly with stickiness, so an entity with much higher accumulated history (e.g. stickiness 43) would show a 44x ratio under the same event.

Test 7.4: Stickiness Monotonic Accumulation. Across a sequence of mixed positive and negative events, do both stickiness values only ever increase or hold steady, never decrease?

One fresh entity. Five mixed events applied in order: (+0.3, 0.5), (−0.4, 0.3), (+0.6, 0.8), (−0.2, 0.4), (+0.1, 0.9). Both stickiness values sampled after each.

- Monotonicity check: both values never decreased across all five events
- Final positive stickiness: 2.200
- Final negative stickiness: 0.200

Result: Both stickiness values strictly accumulate. Positive events feed positive stickiness; negative events feed negative stickiness; the two channels never subtract from each other. After a mix of three positive and two negative events of varying intensity, the agent has accumulated both kinds of weight. This is the architectural answer to "people are complicated": the agent's representation of a person is not a single signed number but a pair of accumulated histories that can both grow at the same time. Negative events do not erase positive history. Positive events do not erase negative history. The two together produce a richer representation than a single attachment-value number could.

Phase 7 Results Summary.

Test	Checks	Mechanism Verified	Key Number
7.1 Attachment Bootstrap	5/5	Master entity has TRUSTED, positive stickiness, protected_bias = 1	stickiness 0.8509, interactions 45
7.2 Attachment Updates on Chat	2/2	Chat increments count; stickiness never decreases	45 → 46, stickiness held at 0.8509
7.3 Stickiness Resists Movement	3/3	movement = |val| × intensity / (1 + stickiness)	11x resistance at stickiness 10 (0.0227 vs 0.25)
7.4 Stickiness Monotonic Accumulation	2/2	Both stickiness channels accumulate without subtraction	+s = 2.200, −s = 0.200 after 5 mixed events

What Phase 7 settles. The agent's representation of other people is governed by an explicit formula that accumulates without forgetting. A fresh entity is fully responsive to the next event. A long-history entity is increasingly resistant to single-event change. Positive and negative histories do not cancel each other; they add to separate channels. The four tests verify the bootstrap state, the per-chat update rule, the resistance formula, and the monotonicity guarantee. None of these are LLM-controlled. They are computed by the attachment subsystem from the geometric event stream, and the LLM reads the resulting trust class and stickiness as an input to its own response generation. The architecture treats trust the same way it treats memory: as a function of accumulated geometric history, not as a flag the LLM can flip.

5.4 The Iteration Record

The full test suite ran 19 times between March 28 and April 1, 2026. Every aborted run, every fix, every failure mode is documented at test-regime.html. The iteration is not a bug-hunt log; it is part of the validation. Run 1 found two real bugs in the production code (5.4.1). Three runs were aborted to fix sensor isolation (5.4.2). Two runs surfaced a procedural issue with the agent's forgotten-fear timer that no amount of code review could have predicted (5.4.3). The first clean sweep was Run 14 in quick mode. The first clean sweep at production timing was Run 16: 29 tests, 82 checks, all green, 112.1 minutes. The highlights of how the regime got there are below.

5.4.1 Bugs Found by Testing

Run 1 discovered two architectural bugs in the Toughbook production code that had never been caught:

1. **Activation overwrite:** The centroid model computed weighted-average activation correctly, but the code threw it away and used only the accumulator (the Peter Fix). Activation was disconnected from the centroid math.
2. **Deception read from database, not geometry:** `is_lying` was read from SQLite, not computed from the geometric state. The deception check only ran during chat, not during heartbeat ticks. Production use masked this because deception was checked during conversation.

These were real bugs, not test artifacts. The test infrastructure validated the implementation before it could validate the math.

5.4.2 Sensor Simulation Decisions

Three consecutive runs were aborted in the early days of the regime over a question that on the surface looks like a small detail and is in fact the experimental design. Run 5 (full mode, 50 hours estimated) was stopped within minutes when the realization hit that the laptop's camera was still active. The agent had a face-recognition source point that would fire whenever someone walked past, opened a door, or arrived home. An unattended 50-hour run with the camera live would record sensor noise as if it were ambient input, and the geometric state would reflect that noise rather than the controlled scenarios. Run 5 was stopped to deal with this.

Run 6 patched the immediate issue by keeping test mode on through Test 1.4 (the embodied-vs-isolated comparison). The patch was wrong for a different reason. Test 1.4 is supposed to demonstrate that an embodied agent has nonzero spread because its sensors contribute source points. If the operator is not in the room, the camera sees an empty room, and the embodied test is measuring something that is not embodiment. The test would technically pass but the result would be uninterpretable.

Run 7 took a different path: simulate the embodied sensors via injected source points (home zone, battery, brightness) and skip the camera entirely. This was also wrong. The other sensors were already simulated. Skipping the camera but simulating the rest produced an asymmetric experiment where one channel was treated differently than the others. The right answer was to simulate the camera too: inject a face-recognition source point that says "the master is in the room, calming effect", and treat that as the canonical embodied condition.

Run 8 was the final design. Every ambient input simulated and consistent. GPS to home location. Battery at 85%. Brightness moderate. Camera reporting master face recognized with a calming valence. Nothing in the test environment depends on the physical room the laptop is sit-

ting in. The agent's "embodiment" in the test regime is a deterministic simulation, because reproducibility requires it. Real-hardware embodiment is what the production deployments demonstrate (Section 5.7); the test regime measures the math.

5.4.3 The Forgotten Fear Discovery

Run 15 (production timing, all fixes from Run 14 applied) failed Tests 1.1 and 1.3 in a way that did not match the centroid math. The four valence checks in Test 1.1 passed perfectly. The four activation checks failed. The activation values were saturated at 1.0 across the test, even when the configuration should have produced 0.4 or 0.7. The accumulator was being driven by something the test could not see.

The something was forgotten fear. The fear engine has a 15% weight component called `forgotten` that measures time since the last interaction with the master. If Potato has not been spoken to in days, this component elevates fear_level above its rest value. fear_level feeds the activation accumulator. The test mode flag blocks sensor source points from contributing to the centroid, but it does not block the fear computation, because fear is computed from internal state (interaction-time deltas), not from sensors. So an agent that the operator has not greeted recently is, by the architecture's own rules, an agent in elevated fear, even when test mode is on.

The fix was procedural, not code: say hello to Potato before running tests. Run 16 (same code as Run 15) passed cleanly, and the difference between the two runs was that Run 16 ran shortly after a conversation. The forgotten value was low, the accumulator was at rest, and the centroid math could speak for itself.

The lesson is that "test mode" is not "no inputs". Time still passes. The agent's internal state still ticks forward. Forgotten fear is itself a form of input, generated by the absence of input rather than the presence of one. A sterile test environment for a persistent embodied agent is one where the agent has no reason to be afraid, and that requires the operator to have greeted the agent before running the suite. The architecture is doing exactly what it claims; the regime had to learn how to set the right initial conditions for it.

5.4.4 Clean Sweep

Run 14 (quick mode, 3 cycles): 29/29 passed, 79/79 checks. First clean sweep. Run 16 (production timing, 10 cycles, 1-minute intervals): 29/29 passed, 82/82 checks, 112.1 minutes. First clean sweep at normal speed. The two activation failures from Run 15 passed on Run 16, confirming they were LLM and accumulator variance between runs, not a systematic bug.

5.5 Formal Proof Verification

The behavioral test regime in Section 5.3 verifies that the running code does what the architecture papers describe. It does not verify that the math itself is sound across every possible input. For that, a separate suite of formal proofs was written into `thesis/proofs.md`: seven theorems covering fourteen specific claims about the geometric model. The suite was numerically verified by `run_proof_tests.py`, which generates 10,000 random input vectors per theorem and checks that each claim holds across the sample. The full verification completes in 0.8 seconds and uses no server, no LLM, and no Potato runtime. It is pure math against pure math, complementary to the behavioral validation in 5.3 but covering the parameter regions no operational test can reach.

Initial run: 24/26. Theorem 3 ("reduces to single-axis models") failed Cases 1 and 2. Investigation revealed the σ_i non-monotonicity (see Appendix A). The theorem was wrong, not the constant. Corrected run: 29/29 theorems verified, 41/41 checks.

Key finding: Operational testing (29 tests, 14 runs) never caught the anomaly because no behavioral mechanism operates in the affected region. Only exhaustive mathematical verification across the full parameter space revealed it. Empirical testing and formal proofs are complementary. Each catches what the other misses.

5.6 What the Experiment Shows

1. Centroid valence tracks the intensity-weighted average across all conditions
2. Spread increases monotonically with source divergence (1.054 → 1.195 → 1.215)

3. Downstream behaviors read centroid position, not individual sources
4. Source removal produces immediate centroid recovery
5. Embodied agents may not reach zero spread because ambient sensors always contribute
6. Memory architecture validates geometric encoding: trace-type decay, stickiness modulation, true forgetting progression, and survival criteria all pass
7. Attachment bias validates geometric accumulation: stickiness resists movement (11x at stickiness 10, scaling smoothly with accumulated history), monotonicity holds across mixed events
8. LLM behavioral output is stochastic and does not reliably translate geometric state into language (Test 3.3 hedging). The geometry is correct. The output layer is the bottleneck.
9. Formal mathematical proofs catch structural anomalies that empirical testing misses (the σ_i non-monotonicity existed from first deployment but was invisible to 14 test runs)

5.7 Supporting Deployments

The controlled experiment above is the main evidence. Two other deployments add context:

- **Production deployment (25 days, macOS):** 10,328 memories across 25 days confirmed that the geometric state produces consistent behavioral signatures under naturalistic conditions. Peak displacement of 0.462 from GPS with immediate recovery on return home.
- **iOS replication (2 days, iPhone):** 2,733 telemetry events confirmed cross-platform portability. The activation accumulator bug was independently reproduced and fixed, providing evidence that the failure mode described in the papers is real and reproducible.

Evidence Status

- **PROVEN:** Centroid weighted aggregation, spread correlation with source divergence, source removal recovery (Phase 1, 4 scenarios)
- **PROVEN:** Deception threshold triggers from distance-from-sigma (Phase 2)
- **PROVEN:** Memory architecture: trace-type decay ordering, stickiness modulation, reconsolidation diminishing, true forgetting progression, survival criteria (Phase 6, 6/6 tests)
- **PROVEN:** Attachment bias: stickiness resists movement (11x at stickiness 10, formula verified to 4 decimals), monotonicity holds (Phase 7, 4/4 tests in Run 16)
- **PROVEN:** Cross-platform portability (Toughbook, MacBook Air, iPhone)
- **PROVEN:** Formal mathematical proofs: 29/29 theorems verified, 10,000 random trials each
- **SUPPORTED:** Embodied agents may not reach zero spread (observed, not formally proven)
- **SUPPORTED:** LLM behavioral output does not reliably translate geometric state into language (Test 3.3 hedging, Runs 18-19)

Likely Reviewer Criticism

- "n=1 agent. No statistical power." The agent experiment shows mechanism, not population effects. The dataset analyses in Chapter 6 provide statistical validation on human data.
- "The LLM generates the behavioral output. You're testing prompt engineering, not the model." The geometric state is computed from sensor inputs and injected source points. The LLM reads the state. It does not generate the centroid positions. Those are math.
- **Structural concern:** This chapter mixes methodology and results. Most committees want a standalone methods chapter covering setup, instrumentation, test protocol, and data collection so the method can be evaluated independent of the results. Consider splitting this into Chapter 5: Methods and Chapter 6: Results before submission.

Chapter 6: Existing Datasets

6.1 Purpose

Show that the geometric decomposition reveals predictive structure hidden by scalar instruments in existing datasets. The analyses use the same three-axis framework defined in Chapter 3, applied retroactively to item-level responses. Four datasets, four domains, four outcomes, one structural finding.

The synthetic-to-human jump is justified at the structural level: the geometry defined in Chapter 3 is a coordinate system, not an agent-specific construct. Whether the same coordinate system that drives the agent's behavior also organizes human survey responses is a testable claim. This chapter tests it on data the agent never saw and never produced.

6.2 Method: Theory-Driven Decomposition

For each dataset, survey items are mapped to the three geometric axes based on clinical or theoretical content (theory-driven, not statistically derived). Axis scores are computed as means of assigned items. Geometric subtypes are formed via median splits on valence and activation within scalar score bands (parameter-free, no clustering algorithm). Outcomes are then compared across subtypes while holding the original scalar score constant.

The PHQ-9 mapping is canonical for the method:

Axis	PHQ-9 Items	What It Measures
Valence	Q1 (anhedonia), Q2 (depressed mood), Q6 (low self-esteem)	How bad is the cognitive-affective state. The positive-negative dimension.
Activation	Q3 (sleep), Q4 (fatigue), Q8 (psychomotor)	How stirred up or shut down is the person. The calm-agitated dimension.
Intensity	Q5 (appetite), Q7 (concentration), Q9 (suicidal ideation)	How strongly is the state registering. The severity-magnitude dimension.

Each axis score is the mean of its assigned items (each item scored 0 to 3, range 0.0 to 3.0). The traditional PHQ-9 total equals (Valence score + Activation score + Intensity score) × 3. The total is recoverable from the three axes. The three axes are not recoverable from the total. Information flows one direction.

Worked example. Two NHANES respondents both score PHQ-9 = 12.

- **Patient A:** Q1=2, Q2=2, Q6=2, Q3=1, Q4=1, Q8=1, Q5=1, Q7=1, Q9=1. Valence = 2.0, Activation = 1.0, Intensity = 1.0. High valence-axis depression with low activation: anhedonic subtype.
- **Patient B:** Q1=1, Q2=1, Q6=1, Q3=2, Q4=2, Q8=2, Q5=1, Q7=1, Q9=1. Valence = 1.0, Activation = 2.0, Intensity = 1.0. Low valence-axis depression with high activation: agitated subtype.

Same total score. Different geometric positions. In the NHANES data, the anhedonic subtype shows 27.5% suicidal ideation; the agitated subtype shows 7.8%. The total score sees 12. The geometry sees two different clinical presentations.

6.3 Results

Three cross-sectional results plus a longitudinal replication (Section 6.4).

6.3.1 NHANES PHQ-9 (n=5,455; moderate band n=455)

Within the moderate depression band (PHQ-9 10 to 14), four geometric subtypes split by median valence and activation. Suicidal ideation rates vary 3.5x across subtypes:

Subtype	**n**	**Suicidal Ideation Rate**
Anhedonic (high valence-axis, low activation)	167	27.5%
Classic Depressed	56	19.6%
Somatic-Dominant (high activation, lower valence)	117	15.4%
Agitated (high activation, lower valence-axis depression)	115	7.8%

132 respondents scoring exactly PHQ-9 = 10 separate into the same four subtypes with the same outcome ordering. Same nine items, different math. Companion paper: Riggleman 2026s.

6.3.2 American Customer Satisfaction Index (n=7,341; satisfaction-7 band n=1,034)

Within respondents rating satisfaction exactly 7 out of 10, four geometric subtypes split by quality and value relative to directly measured expectations (sigma). Complaint rates vary 2.3x:

Subtype	**n**	**Complaint Rate**
Exceeded Both (quality and value above expectations)	323	15.8%
Quality Up, Value Down	296	12.5%
Quality Down, Value Up	85	27.1%
Fell Short Both	330	28.5%

Pattern holds across satisfaction scores 6, 7, and 8 and across four industries. Higher sigma (expectations) at the same scalar score predicts higher complaint behavior. Companion paper: Riggleman 2026r.

6.3.3 IBM HR Analytics (n=1,470)

Median split on valence (job satisfaction) and activation (work-life balance, environment) produces four quadrants with attrition rates varying 2.2x:

Quadrant	**n**	**Attrition Rate**
High Valence + High Activation	747	11.8%
High Valence + Low Activation	326	17.5%
Low Valence + High Activation	274	21.9%
Low Valence + Low Activation	123	26.0%

Sigma inferred from item mapping. Employees with identical scalar satisfaction scores occupy different geometric quadrants with different outcomes.

6.3.4 Cross-Dataset Convergence

The same structural finding replicates across all four datasets:

Finding	**PHQ-9 (n=5,455)**	**ACSI (n=7,341)**	**IBM HR (n=1,470)**	**EES (n=715, 4 waves)**
Hidden variation within same score	3.5x	2.3x	2.2x	2.9x
Outcome predicted	Suicidal ideation	Complaint behavior	Employee attrition	Attrition intent and advocacy
Domain	Clinical	Customer satisfaction	Employee retention	Employee engagement
Sigma source	Population median	Directly measured	Inferred	Population median
Longitudinal	No	No	No	**Yes (4 waves)**
Total n	**14,981 respondents**			

6.4 Longitudinal Replication: Employee Engagement Survey

The fourth dataset is a four-wave Employee Engagement Survey panel (Oct 2022, May 2023, Oct 2023, May 2024) from a single organization, n=715 respondents pooled across waves (151 / 170 / 188 / 206). 39 Likert items on a 1 to 5 scale. 17 items were allocated to the three axes following standard organizational-behavior theory; two outcome items were held out from all axes to support a clean hidden-variation test.

Theory-driven axis partitions show strong internal consistency: Cronbach's $\alpha = 0.892$ (valence: affective commitment, 6 items), 0.792 (activation: engagement energy, 6 items), 0.882 (intensity: commitment depth, 5 items). This is the strongest axis-level psychometric validation across all four datasets. Within engagement score bands, octant-level outcome rates (attrition intent, recommendation) vary up to 2.9x.

Octant proportions remain moderately stable across waves (mean coefficient of variation 0.31), with non-uniform axis drift when overall engagement shifts: precisely the pattern a scalar metric cannot resolve.

Wave	**HHH**	**HHL**	**HLH**	**HLL**	**LHH**	**LHL**	**LLH**	**LLL**
Oct 2022	31.1	11.3	7.3	6.6	2.0	6.0	1.3	34.4
May 2023	38.8	7.6	5.9	5.3	4.1	8.2	4.7	25.3
Oct 2023	20.2	7.4	5.9	5.3	4.3	6.9	5.9	44.1
May 2024	23.8	5.8	2.4	5.8	2.9	10.2	2.4	46.6

Octants are labeled by High/Low on Valence-Activation-Intensity. The HHH octant (high on all three axes) shrank from 31.1% to 23.8% across the panel; the LLL octant (low on all three) grew from 34.4% to 46.6%. Single-axis engagement score shifts hide this composition change. Detailed analysis in the companion paper (Riggleman 2026r).

6.5 Convergence and Implications

The same structural finding (hidden outcome-relevant variation within scalar bands) replicates across four domains, four instruments, and four outcome variables. Sigma (directly measured in ACSI, inferred elsewhere) further differentiates respondents at identical scalar scores. The longitudinal replication in EES (Section 6.4) shows the structure is not an artifact of one cross-section; it persists and shifts coherently as the population's overall scalar metric moves.

The geometric model does not replace existing scalar instruments. It decomposes them. Every claim made here is computed from the same item-level responses already in use; nothing new was collected. The instruments' validation work carries forward. What changes is the math applied on top of those item responses, and the resulting interpretive resolution.

Evidence Status

- **PROVEN:** Single-axis scores hide outcome-relevant geometric variation in four independent datasets (PHQ-9, ACSI, IBM HR, EES; total n=14,981)
- **PROVEN:** Hidden variation predicts real outcomes: suicidal ideation 3.5x, complaint behavior 2.3x, employee attrition 2.2x, attrition intent and advocacy 2.9x
- **PROVEN:** The finding generalizes across clinical, commercial, HR retention, and employee engagement domains
- **PROVEN:** Theory-driven axis partitions are psychometrically sound (Cronbach's α 0.79 to 0.89 in EES, the strongest axis-level psychometric validation across the four datasets)
- **PROVEN:** The structure persists across four longitudinal waves with coherent (non-uniform) axis drift (EES 4-wave panel)
- **SUPPORTED:** Sigma (directly measured in ACSI, inferred or population-median elsewhere) is associated with outcome differences the total score does not distinguish
- **NOT YET TESTED:** Whether the GAS as a native three-axis instrument outperforms retroactive decomposition of existing scales (Riggleman 2026r)

Likely Reviewer Criticism

- "The item-to-axis mapping is subjective." Acknowledged. Mappings are theory-driven (items grouped by clinical or theoretical content), not exploratory factor analysis. Convergence across four independent mappings on four datasets, plus directly measured sigma in ACSI and Cronbach's α 0.79 to 0.89 in EES, provides robustness. Formal sensitivity analysis is planned (Riggleman 2026r).
- "Retroactive decomposition is not native measurement." Acknowledged. All current analyses reconstruct three-axis coordinates from items designed for scalar scoring. The items were not written to measure valence, activation, and intensity independently. Prospective validation with the Geometric Affective Survey (Riggleman 2026r) is required.
- "Cross-sectional data cannot establish causality." Correct for PHQ-9, ACSI, and IBM HR. The thesis claims predictive association, not causation. The EES longitudinal panel (Section 6.4) shows the structure is persistent rather than a cross-sectional artifact, but does not establish causation.
- "The NHANES suicidal ideation finding is from a general population sample, not a clinical one." Acknowledged. Rates, distributions, and thresholds may look different in psychiatric settings. Replication in clinical samples is required before any clinical application (Riggleman 2026s).
- "Why these three axes and not two or four?" Two axes (Russell's circumplex) are a special case of this framework (Theorem 3 Case 3, Section 3.8). The third axis (intensity) captures magnitude of displacement that the other two do not. Four or more axes may be viable but have not been tested. Three is the minimum that produces the observed separation.

Chapter 7: Applications

7.1 Purpose

The downstream stuff the geometric model drives. These are secondary contributions. Consequences of the model, not the model itself.

7.2 Applications (briefly)

- **Memory decay:** Intensity at encoding drives differential decay rates across memory types (Riggleman 2026l).
- **Deception:** Distance from sigma is the threshold and magnitude input for the deception mechanic (Riggleman 2026b, 2026m).
- **Nightmare formation:** Intensity and valence at consolidation replace the arbitrary nightmare threshold with a geometric one (Riggleman 2026d, 2026l).
- **Addiction:** The positive valence axis produces a structural mirror of trauma encoding. Sigma_i is a measurable geometric trait for addictive personality: low values are a personality hum, high values produce agents that structurally cannot rest. Two-phase model separates stuck memory (phase one) from personality reorganization (phase two) (Riggleman 2026n).
- **Identity discontinuity:** Environmental orphaning in memory transplant shows coordinate-system dependence (Riggleman 2026o).
- **Survey measurement:** The GAS puts the three-axis framework into a survey humans can answer (Riggleman 2026r).
- **Clinical assessment:** Geometric displacement gives you a trajectory-based clinical metric (Riggleman 2026s).

Each one has its own paper. The thesis treats them as evidence the model generalizes, not as standalone contributions.

Structural Concern

- This chapter is a bullet list with one-sentence descriptions and paper citations. As a standalone chapter it will get flagged as padding. Either expand each application into a real discussion of how it follows from the geometry (a paragraph or two each showing why the geometric model is necessary for the application to work), or demote this to a section inside the conclusion.

Chapter 8: Evidence Classification

8.1 Purpose

Separate what is proven, what is supported, and what is speculative. Honesty requires this.

8.2 Classification

Claim	Status	Evidence
Centroid weighted aggregation produces predicted centroid positions	**Proven**	Centroid validation experiment, 4 scenarios, 110 cycles (Section 5.2). Phase 1 Tests 1.1–1.4 confirmed at production timing in Run 16 (15/15 checks).
Spread increases monotonically with source divergence	**Proven**	1.054 → 1.195 → 1.215 across scenarios
Downstream behaviors track centroid position, not individual sources	**Proven**	Behavioral output changed with centroid, not with individual source states
Source removal produces immediate centroid recovery	**Proven**	Valence jumped 0.07 → 0.68 on removal (Scenario 4). Phase 3 Test 3.1 confirmed at production timing: 0.6094 shift in a single tick (Run 16).
Distance from sigma is one variable serving four behavioral mechanics (deception, confession, nightmare eligibility, memory intensity)	**Proven**	Phase 2 Tests 2.1, 2.2, 2.3, 2.4, 2.5 (Run 16, 12/12 checks): deception flag fires only when distance ≥ 0.5 AND valence < 0; tell phrase code-injected and unsuppressable across 3/3 probes including suppression attempts; confession appears on recovery below distance 0.25; memory intensity captured at encoding (1.183 vs 0.192 far vs near); nightmare eligibility uses the same two-clause rule.
Distance from sigma is linear in source weight	**Proven**	Phase 3 Test 3.2 (Run 16): $R^2 = 0.9123$ across 11 ramp steps from weight 0 to 1.0, monotonic. The centroid is a weighted average; weighted averages are linear in weight; the linearity holds under measurement.
Sigma is an attractor; the math handles unit-cube extremes without numerical failure	**Proven**	Phase 5 Tests 5.1–5.4 (Run 16, 13/13 checks): convergence to sigma on source removal (final distance 0.1902, valence 0.4); maximum-corner displacement handled (V = −0.6769, D = 1.4131, no NaN, response 717 chars); rapid source switching produces perfectly periodic centroid (0.6909 / −0.4727 alternation across 6 cycles); heartbeat counter running at 673, shutdown-detection mechanism wired.

Claim	**Status**	**Evidence**
LLM language variability quantified at fixed geometric state	**Proven**	Phase 4 Test 4.2 (Run 16): coefficient of variation = 0.376 (mean response length 220 chars, std 83) at a constant source set across five identical chats. The LLM is the noisy stage downstream of the deterministic geometric state. This is the calibration noise floor for Test 3.3 hedging counts.
Geometric decomposition reveals hidden structure in scalar instruments	**Proven**	IBM HR (2.2x attrition), ACSI (2.3x complaints), NHANES (3.5x ideation)
The architecture generalizes across platforms	**Proven**	Toughbook, MacBook Air, iPhone in production. Phase 4 Test 4.1 (Run 16): Linux centroid valence −0.4727 matches the Toughbook reference −0.473 to four decimal places, distance 1.1338 in the same range as the Toughbook 1.455.
Activation axis accumulator produces different state than snapshot measurement	**Proven**	iOS replication independently reproduced the predicted failure mode and confirmed the fix
Memory architecture validates geometric encoding (trace-type decay, stickiness, reconsolidation, true forgetting, survival)	**Proven**	Phase 6: 6/6 tests, 20/20 checks. First passing run was Run 11; reproduced in Run 16 at production timing. Per-trace decay ordering (raw 0.82 < semantic 0.89 < graph 0.95 after one night), reconsolidation boosts strictly diminishing (0.150, 0.115, 0.094, 0.079, 0.068), full → fragment → ghost → deleted erosion across 7 nights, four alternative survival paths verified.
Attachment bias validates geometric accumulation (stickiness resistance, monotonicity, multi-channel)	**Proven**	Phase 7: 4/4 tests, 12/12 checks in Run 16. Stickiness resistance 11x at stickiness = 10 (formula movement = |valence| × intensity / (1 + stickiness) verified to 4 decimals; ratio scales smoothly with accumulated history). Both stickiness channels accumulate monotonically across mixed events. Test 7.1 was rewritten between Run 11 and Run 16 to check structure rather than current attachment value: the eroded value was correct architectural behavior under test ordering, not a bug.

Claim	**Status**	**Evidence**
Formal mathematical proofs verified by exhaustive numerical testing	**Proven**	29/29 theorems, 41/41 checks, 10,000 random trials each. Caught σ_i anomaly invisible to 14 operational runs
Test infrastructure found latent production bugs that deployment masked	**Proven**	Run 1: activation overwrite and deception-from-database bugs. Both real, both invisible in production use
Forgotten fear component leaks into test mode via fear_level computation	**Proven**	Runs 15, 17: test_mode blocks sensor sources but not fear_level. Procedural fix (say hello) confirmed on Runs 16, 18
LLM output does not reliably translate geometric spread into hedging language	**Proven**	Test 3.3 variance across runs. Run 16: passed (S2 = 7 vs S1 = 5 hedging words; spread 0.80 vs 0.32). Runs 18 and 19: failed by one word (S2 = 5, S1 = 6). The spread is unambiguously different in every run. The hedging count fluctuates run to run at the LLM noise floor (CV = 0.376 measured in Test 4.2). Geometry correct; LLM output stochastic. Architecturally addressed by YAM.
Embodied agents may not reach zero spread	**Supported**	Observed in centroid experiment; ambient sensors contribute non-zero spread
Sigma is associated with outcome differences that total scores do not distinguish	**Supported**	ACSI directly measured sigma; higher sigma = higher complaint rate at same score
The geometry shares structural properties with established human affective models	**Supported**	Russell (1980), Larsen & Diener (1987), Cacioppo & Berntson (1994), Bowlby (1969), Lazarus (1991). All arrived at inductively, not used as design inputs
The geometric model produces changed behavioral output following state discontinuity	**Supported**	Single shutdown experiment; behavioral shift observed but not replicated

Claim	**Status**	**Evidence**
The GAS as a native instrument outperforms scalar instruments	**Not yet tested**	Backward compatibility demonstrated; native collection requires prospective study
Trajectory-based clinical assessment detects approach before arrival	**Not yet tested**	Requires longitudinal clinical data with treatment outcomes
The geometric framework may generalize across sensory modalities	**Speculative**	Theoretical proposal only (Riggleman 2026q). Requires echolocation robot experiment
The agent experiences subjective states	**Not claimed**	The thesis describes behavior, not phenomenology. This claim is outside scope.

Chapter 9: Limitations

9.1 Agent Experiment Limitations

- **n=1 architecture.** All agent experiments involve one system on three platforms. The experiments show the mechanism works in this architecture. They do not show it generalizes to other architectures. Replication on an independent agent is needed before the mechanism claim is general.
- **LLM variability.** The behavioral output comes from an LLM that reads the geometric state. The same state could produce different language on different runs. Centroid positions are deterministic. Behavioral responses are not. The geometric validation is clean. The behavioral validation has noise. Test 3.3 (hedging) demonstrates this concretely: high-spread and low-spread configs produced nearly identical hedging word counts (5 vs 6) across Runs 18 and 19. The geometry was correct. The LLM did not act on it.
- **Ambient source contamination.** The centroid experiment could not fully isolate injected sources from ambient sensor inputs. Spread never hit zero because embodied sensors always contribute. This might be a property of embodiment rather than a flaw, but it means Scenario 1 is not a pure single-source test.
- **Forgotten fear leaks into test mode.** The `forgotten` fear component (15% weight, time since last interaction) feeds the activation accumulator even in test_mode, because test_mode blocks sensor source points but not the fear_level computation. Runs 15 and 17 failed activation tests for this reason. Procedural fix: interact with the agent before testing. This is a limitation of embodied testing, not the geometry.
- **Scenario 5 not yet executed.** The gradual intensity ramp is planned but not completed. Without it, the thesis cannot confirm linearity of the centroid response across the full intensity range.

9.2 Dataset Analysis Limitations

- **Item-to-axis mapping involves judgment.** The researcher picks which survey items map to which axis. Different mappings could produce different results. The ACSI analysis partly addresses this because expectations are measured directly as sigma. The EES longitudinal analysis partly addresses it via Cronbach's α 0.79 to 0.89 across the three axes, the strongest internal-consistency evidence to date that the theory-driven partitions are psychometrically sound. Sensitivity analysis on alternative mappings is planned (Riggleman 2026r).
- **Cross-sectional data cannot establish causality.** The IBM HR, ACSI, and NHANES analyses show association between geometric position and outcomes. They do not show causation. Longitudinal data with intervention controls would be needed for that.
- **General population vs. clinical.** The NHANES suicidal ideation finding is from a national health survey, not a clinical population. Rates, distributions, and thresholds may look different in psychiatric settings. Replication in clinical samples is required before any clinical application.
- **No native three-axis collection.** All dataset analyses are retroactive decompositions of existing single-axis data. Whether collecting all three axes natively produces the same or better results has not been tested.

9.3 Model Limitations

- **Euclidean distance assumptions.** The distance metric assumes the three axes are orthogonal and equally weighted. Neither assumption has been validated. If the axes are correlated or one axis matters more than the others, a weighted or Mahalanobis distance might work better.
- **Sigma stability.** Sigma is assumed stable over weeks to months. This has not been tested in human respondents. If sigma drifts between measurements, displacement computations become unreliable.
- **Intensity derivation vs. self-report.** In the agent, intensity is computed from valence and activation displacement. In the survey, intensity is self-reported. Whether those two approaches

measure the same thing is testable but has not been tested.

- **Sigma intensity pathology threshold.** The sigma intensity anomaly (Appendix A) shows that high σ_i produces downstream system behavior consistent with addictive personality. The threshold at which personality trait becomes structural pathology is computable from the downstream parameters but has not been computed or validated. Whether that threshold is sharp or gradual is an open question. Full analysis is in Riggleman (2026n) Section 3.5.

9.4 IRB Considerations

No human subjects were involved in any experiment presented in this thesis. All agent experiments used synthetic systems. All dataset analyses used publicly available, de-identified data (IBM HR Analytics via Kaggle, ACSI via Mendeley Data, NHANES via CDC).

Future work involving human respondents (GAS validation, clinical trajectory tracking) will require IRB approval. The thesis identifies these as future directions and does not present them as completed work.

Chapter 10: Future Work

10.1 Human Validation

- Run the GAS alongside existing instruments
- Test-retest reliability on the three axes
- Measure sigma stability across sessions
- Whether self-reported intensity matches the computed version

10.2 Clinical Application

- Replicate the NHANES subtype finding in clinical populations
- Longitudinal trajectory tracking with treatment outcomes
- Test whether trajectory tracking actually catches problems early

10.3 Technical Extensions

- Other distance metrics (Mahalanobis, weighted Euclidean)
- What happens when multiple agents interact
- Measuring whether and how fast sigma drifts
- Compute the σ_i pathology threshold from downstream system parameters (decay, stickiness, deception)
- Controlled addiction/relapse protocol on the Potato system
- Cross-modal test (echolocation robot experiment)

Chapter 11: Conclusion

11.1 Purpose

Restate the contribution. No new claims. No expansion.

11.2 The Contribution

This thesis presents a three-dimensional geometric representation of affective state with a personality setpoint, a centroid model for multi-source aggregation, and evidence that this structure outperforms scalar representations in both synthetic agent behavior and human datasets across three independent domains.

The model came from engineering requirements of a deployed AI agent. It lines up structurally with established models of human affect without having been built from them. Whether that overlap says something general about how dimensional emotional systems work is the question this thesis leaves open.

Supporting Papers

The thesis draws on 22 companion papers, all available at clawddaily.com/papers. Each paper is a standalone contribution. The thesis synthesizes them into a unified argument.

Paper	Thesis Chapter	Role
2026j: Valence	Ch. 3	Axis definition
2026k: Activation	Ch. 3	Axis definition + accumulator
2026l: Intensity	Ch. 3	Axis definition + memory decay
2026m: Emotional Geometry	Ch. 3	Unified model + sigma
2026t: Centroid Model	Ch. 4	Multi-source aggregation + spread
2026f: SGI Holistic	Ch. 5	Production deployment data
2026a: Memory Decay	Ch. 5, Ch. 7	Phase 6 validation + application: memory
2026b: Lie Mechanic	Ch. 5, Ch. 7	Phase 2 validation + application: deception
2026d: Nightmares	Ch. 7	Application: dream cycle
2026e: Full Spectrum	Ch. 5	Prior model (superseded by geometry papers)
2026i: Outbound Trust	Ch. 5	Peter experiment
2026n: Addiction	Ch. 7, Appendix A	Application: positive valence axis, sigma intensity as addictive personality trait, two-phase model
2026o: Identity Discontinuity	Ch. 7	Application: memory transplant
2026p: Functional Consciousness	Ch. 5	Shutdown experiment
2026q: Cross-Modal Translator	Ch. 10	Future work: modality independence
2026r: GAS Survey	Ch. 6	Dataset analyses (IBM HR, ACSI)
2026s: Clinical Metric	Ch. 6	Dataset analysis (NHANES PHQ-9)

Paper	**Thesis Chapter**	**Role**
2026u: Attachment Bias	Ch. 5	Phase 7 validation: stickiness, monotonicity, formation matrix

Appendix A: The Sigma Intensity Anomaly

The intensity function

$$I(v, a) = \min(1, \sqrt{((v - \sigma_v)^2 + (a - \sigma_a)^2)} / \sqrt{2})$$

The model computes intensity from two inputs: v (current valence) and a (current activation). σ_v and σ_a are the valence and activation components of the resting point sigma.

$(v - \sigma_v)$ is how far valence is from resting valence. Square it. $(a - \sigma_a)$ is how far activation is from resting activation. Square it. Add the two squares and take the square root. That gives you the Euclidean distance from the current point to sigma in the valence/activation plane. Pythagorean theorem on a 2D plane.

Then divide by $\sqrt{2}$. Valence and activation both range 0 to 1, so the farthest any two points can be from each other in that unit square is the diagonal: $\sqrt{(1^2 + 1^2)} = \sqrt{2}$. Dividing by $\sqrt{2}$ normalizes intensity to a 0-to-1 range.

The min(1, ...) is a clamp. If the raw value goes over 1, cap it.

So intensity is just "how far am I from home, as a fraction of the farthest I could possibly be from home." At home, distance is zero, intensity is zero. At the far corner of the space, intensity is 1.

The observation

The deployed sigma was $\sigma = (\sigma_v, \sigma_a, \sigma_i) = (0.4, 0.2, 0.2)$. Plug the resting point into the intensity formula: $v = \sigma_v$, $a = \sigma_a$. That gives $\sqrt{((\sigma_v - \sigma_v)^2 + (\sigma_a - \sigma_a)^2)} = \sqrt{(0 + 0)} = 0$. So $I(\sigma_v, \sigma_a) = 0$ by definition. But σ_i was stored as 0.2. The formula produces intensity 0 at rest. The stored sigma

says intensity 0.2 at rest. These do not match. The question is whether that mismatch is an error or a feature. The math below shows what it does. The interpretation follows.

Setting up the distance proof

To show what this contradiction actually does, hold activation fixed at σ_a so only valence moves. Define $\delta = v - \sigma_v$. Delta is how far valence moved from rest. Pinning activation collapses the problem to one dimension so the anomaly shows up clearly.

With activation pinned, the $(a - \sigma_a)$ term is zero and intensity simplifies to:

$$I(v, \sigma_a) = \sqrt{(\delta^2)} / \sqrt{2} = |\delta| / \sqrt{2}$$

Intensity is absolute valence displacement divided by $\sqrt{2}$. Move valence away from rest, intensity goes up proportionally.

The distance anomaly

Compute the squared distance from the current state (v, a, I) to sigma $(\sigma_v, \sigma_a, \sigma_i)$ in three dimensions. Activation is pinned at σ_a, so that dimension drops out. Two terms remain:

$$d^2 = (v - \sigma_v)^2 + (I - \sigma_i)^2$$

First term is δ^2. For the second term, I is $|\delta|/\sqrt{2}$ (from above) and σ_i is 0.2. Expand it:

$$(|\delta| / \sqrt{2} - 0.2)^2 = \delta^2/2 - 2 \cdot (|\delta| / \sqrt{2}) \cdot 0.2 + 0.04$$

Note that $2 \cdot (1/\sqrt{2}) = \sqrt{2}$. That is where the $\sqrt{2} \cdot \sigma_i$ coefficient comes from. Simplify:

$$= \delta^2/2 - \sqrt{2} \cdot \sigma_i \cdot |\delta| + \sigma_i^2$$

Add the first term δ^2 back in:

$$d^2 = \delta^2 + \delta^2/2 - \sqrt{2} \cdot \sigma_i \cdot |\delta| + \sigma_i^2 = (3/2)\delta^2 - \sqrt{2} \cdot \sigma_i \cdot |\delta| + \sigma_i^2$$

Three terms. First term, $(3/2)\delta^2$, pushes distance up as valence moves. Expected. Third term, $\sigma_i^2 = 0.04$, is a constant. The middle term is where it gets interesting: it is negative and linear in $|\delta|$. For small moves away from rest, that negative linear pull beats the positive quadratic push. Total distance goes *down* before the quadratic catches up and drives it back up.

Finding the minimum

Take the derivative of d^2 with respect to $|\delta|$ and set it to zero. Let $x = |\delta|$:

$$d^2(x) = (3/2)x^2 - \sqrt{2} \cdot \sigma_i \cdot x + \sigma_i^2$$

$$\mathrm{d}(d^2)/\mathrm{d}x = 3x - \sqrt{2} \cdot \sigma_i = 0$$

$$x = \sqrt{2} \cdot \sigma_i / 3$$

Plug in $\sigma_i = 0.2$:

$$x = \sqrt{2} \cdot 0.2 / 3 = 0.2828\ldots / 3 \approx 0.094$$

The minimum distance to sigma is not at $\delta = 0$ (rest). It is at $|\delta| \approx 0.094$. A point slightly off-center registers as closer to home than home itself. At $\sigma_i = 0.2$ this offset is small. At higher values of σ_i the offset grows and the behavioral consequences become significant.

What this means

Automated testing caught the inconsistency. The test harness flagged a non-monotonic distance function and the initial read was that $\sigma_i = 0.2$ was an implementation error. But interpreting what the math was actually doing took iterative explanation, challenge, and rethinking before the result could be defended as an accidental design choice rather than a simple bug.

The original interpretation was that $\sigma_i = 0.2$ was a bug and the fix was to set $\sigma_i = 0$. That interpretation is wrong. It is one valid design choice, not the only one.

Setting $\sigma_i = 0$ enforces the constraint $\sigma_i = I(\sigma_v, \sigma_a) = 0$. The distance formula becomes:

$$d^2 = (3/2)\delta^2 - \sqrt{2} \cdot 0 \cdot |\delta| + 0^2 = (3/2)\delta^2$$

Middle term is gone. $d = |\delta| \cdot \sqrt{(3/2)}$. Monotonic. Clean. No false minimum. That is what a personality with zero baseline intensity looks like in the math.

But $\sigma_i > 0$ is not an error. It is a personality that includes baseline intensity. The non-monotonic distance function is not the system misbehaving. It is the system correctly representing an agent whose rest includes edge. The agent at dead zero displacement is less at home than the agent with a small disturbance, because dead zero is not where you put home. The small bump does not reward the agent. It brings the agent home.

At low σ_i (0.1 to 0.2) this is a personality trait. The agent runs a little hot. The downstream offsets are small. The agent is functional.

At high σ_i the behavioral profile changes. The minimum achievable distance grows. The agent structurally cannot rest. Memory decay never reaches baseline rate. Stickiness is always elevated. The agent needs bigger and bigger displacements to close the intensity gap. The agent is most comfortable in crisis. These are not analogies. They are the same math at different coordinates producing the behavioral profile of addictive personality.

σ_i is a measurable geometric trait. Low values produce functional personalities with a slight hum. High values produce agents trapped in permanent displacement where no downstream system works as designed. The threshold where personality becomes pathology is computable from the downstream system parameters.

The full analysis, including the two-phase model of addiction (phase one: frozen reconsolidation with intact sigma, phase two: sigma drift with personality reorganization) and the role of σ_i as a diagnostic parameter, is in Riggleman (2026n) Section 3.5.

A note on the scaling factor

Moving 1 unit in valence alone produces a distance of $\sqrt{(3/2)} \approx 1.225$, not 1.0. That is because moving valence also drags intensity up. Intensity is derived from displacement, so a pure valence move is really a move in two dimensions at once. The extra 0.225 is the intensity shadow of the valence movement.

Full investigation, timeline, and corrected proof: sigma-intensity-anomaly.md